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DELSU Post UTME Past Questions & Answers



**POST JAMB SCREENING TEST
2018/2019 SESSION DELSU**

ENGLISH

In each of the questions choose the option nearest in meaning to the word or phrase underlined

1. When a man is immune to an illness, he is

- W. Opposed to it X. Addicted to it
Y. Attached to it Z. Protected against it.

2. Sitting majestically on his throne is the Ovie of Agbarho flanked by some of his wives W. Surrounded X. Guarded
Y. Protected Z. Neglected

3. Ese is an ardent supporter of technical education
W. Cogent Y. An ignorant
X. A passionate Z. An optimistic

4. Sacam Hasseen's epitaph has been demolished.
W. Monument X. Embodiment
Y. Book Z. Farmland

5. There is obvious symmetry in the whole presentation W. Orderliness X. Hesitation
Y. Excitement Z. Confusion

6. The state government appointed a commission of inquiry to go into the community's complaints carefully and without prejudice W. Account for
X. Search
Y. Investigate Z. None of the above

7. As he was a gullible leader, his followers took Advantage
W. He was slow to act
X. He was simple minded to a fault
Y. He was partial and unfair in dispensing justice
Z. He was weak and unable to enforce his Authority

8. He has a big heart but he is inept at following a witty conversation
W. Large X. Huge
Y. Warm Z. Placid

In each questions complete each sentence by choosing the option that most suitably fills the space.

9. The managing director did not pay his last month _____
W. Didn't he X. Had he not
Y. Is not it Z. Did he?

10. It was a long poem but he learnt it
W. Off head X. From memory
Y. By heart Z. In memory

11. The traditional folktales form an _____ to the book
W. Appendix X. Overview
Y. Insertion Z. Index

12. _____ marked an important step in the history of the industrial revolution
W. The process of locomotion
X. The discovery of the engine
Y. The discovery of locomotion
Z. The invention of the locomotive engine

Read the passage below and answer questions 13-17

Word Englishes point out the present day global spread of English and the variant roles of the language in different societies. Nearly every speaker of English today has been exposed to different varieties of global English. English is no longer just 'one language'. It comes in many different shapes and sizes as it were. It is quite different in many counties and localities where it has been adopted. To grasp this phenomenon linguists have come to talk of different Englishes. One of the main reasons why English is learnt in developing countries by people from all works of life is the belief that it is a primary gateway for better jobs and a better life. English is undergoing indigenization and nativization in Anglophone African countries. In many countries of Africa and Asia, there are now asymmetrical bilingual children since they grow up speaking English as the first and most important language. Some have experienced language loss in their indigenous language.

13. Indigenization and Nativization of English as used in the passage means
W. Speaking English as a native and indigenous language
X. Speaking English as a mother tongue
Y. Speaking different varieties of English
Z. Adding features of indigenous language to English

14. According to the author "asymmetrical bilingual children are:
W. Those who speak English and their mother tongue fluently
X. Children who learn their mother tongue before learning English
Y. Children who can only speak English
Z. Those who learn to speak English before learning their mother tongue.

15. The adverse effect of English on children in Anglophone African countries is that
 W. Some children speak pidgin instead of standard English
 X. Some children cannot speak their mother tongue
 Y. Some children focus mainly on learning English at school
 Z. Parents encourage their children to speak Only English
16. Which of these statement does not express the opinion of the author?
 W. Many people in Africa and Asia learn English due to its economic value
 X. English is a language devoid of varieties
 Y. English is learnt by many people in Developing countries
 Z. English is not a mother tongue of many of its speakers
17. In the writer's opinion, English
 W. Has the highest number of speakers in the world
 X. Is spoken only in Asia and Africa
 Y. Has little economic value
 Z. Is the most important language in the World

In each question choose the most appropriate option opposite in meaning to the word(s) underlined

18. The explosive growth of world population has not been caused by a sudden increase in human fertility
 W. Fantastic X. Gradual
 Y. Combustible Z. Dangerous
19. Unless we see the evils clearly, contending against them is like fighting the air
 W. Avoiding X. Struggling with
 Y. Combating Z. Repelling
20. My sister's condition is appalling
 W. Pleasant X. Complex
 Y. Cloudy Z. Simple
21. This is a heinous act W. Laudable Y. Deplorable
 Y. Forgettable Z. Indelible

22. The long straight trunk of the tree is ideal for roofing.
 W. Perfect X. Best
 Y. Most unsuitable Z. Poor
23. The examiner was most flexible in his instruction
 W. Rigid Y. Correct
 X. Stiff Z. Upright
24. The discussion become animated
 W. Lively X. Intellectual
 Y. Unruly Z. Specialized
25. I am always hard up before the end of each month
 W. In no need X. Broke
 Y. Upset Z. Borrowing

APTITUDE TEST

Identify the ODD one in the options for each number

26. W. Bostwana X. Kenya
 Y. Egypt Z. Tunisia
27. W. Yahoo X. Hotmail
 Y. Gmail Z. E-mail
28. W. Kwara X. Lokoja
 Y. Benin City Z. Umuahia
29. W. NECO X. WAEC
 Y. NABTEB Z. JAMB
30. W. Yen X. Pounds
 Y. Naira Z. Dollar

Note 4A =15, 6A7=35 and 9AI = 8. Use this logic to find the missing number in questions 31-35

31. The present secretary General of United Nations is
 W. Kofi Anan X. Bankimoon
 Y. Hillary Clinton Z. Barrack Obama
32. The slogan 'The Big Heart' applies to state
 W. Edo X. Anambra
 Y. Kano Z. Delta
33. $13A14 =$
 W. 169 X. 168
 Y. 182 Z. 156

34. (5 A []) A5 = 75
W. 4 X. 2
Y. 3 Z. 16

35. $2A(7A5) = []$
W. 70 X. 55
Y. 30 Z. 65

Select the correct simile that best completes each expression below. 36. As clear as _____ W. White
X. Glass

Y. Crystal Z. Snow

37. As proud as a _____ W. Peacock X. Prince
Y. King Z. Star

38. As blind as a _____
W. Toad X. Bat
Y. Cockroach Z. Rock

For each number below (39 & 40) identify the anagram to the given word

39. EARTH W. Planet X. Ear
Y. Heart Z. Universe

40. DIET
W. Food X. Die
Y. Tied Z. Nutrition

If 8mmm 6 =16, 10mm 5=25, 1mmmmm 20 = 4 evaluate X if

41. 6mmm 4 = X
W. 24 X. 8
Y. 12 Z. 6

42. Xmm3 = 24
W. 48 X. 9
Y. 18 Z. 16

43. $\frac{1}{4}$ mmm 24 = x
W. 3 X. 6
Y. 6 Z. 2

Choose the word that best completes the following phrase

44. Lion is to roar as elephant is to _____
W. Trumpet X. Rumble
Y. Boom Z. Bleat
45. Flock is to bird as is to Lion
W. Group X. Gang
Y. Pride Z. Fleet

GENERAL PA PER

46. Which of the following statement is true?
W. The volumes of similar solids are in the ratio of the cube of their line at dimension
X. The sum of angles in a triangle is four right Angles
Y. If two un-parallel lines are transverse by a third line, then the corresponding angles must Be equal.
Z. Hero's formula is used for computing Area of a triangles when only one of it's side is given.

47. Given that the matrix $\begin{bmatrix} 3 & 6 \\ 4 & x \end{bmatrix}$ is singular, then the value of x is
W. 10 X. 8
Y. 12 Z. 14

48. Find the area of the curved surface of a cone whose base radius is 3cm and whose height is 4cm in term of π

49. Find the LCM of $x^2 + x$, $x^2 - 1$, $x^2 - x$ W. $x(x+1)(x-2)$
X. $x(x-1)(x+1)$
Y. $(x^2 + 1)(x+2)$ Z. $(x+1)x^2$

50. Suppose that $\log y + 3\log x = 2$, express y in terms of x
W. $y = 2x^2$ X. $y = 52x^2$
Y. $\frac{=100}{x^3}$ Z. $y = \frac{x^3}{54}$

51. The quadratic equation whose roots are 4 and -7 is
W. $x^2+3+28 = 0$ X. $x^2-3x+28 = 0$
Y. $x^2-3x-28=0$ Z. $x^2+3x-28=0$

52. The remainder when $f(x) = x^3 - 3x^2 + x - 5$ is divided by $x + 2$ is
W. -18 X. -22
Y. -27 Z. -32

53. Solve the equation $(0.3)^{x+1} = (0.7)^x$
W. -2.316 X. -3.112
Y. -1.421 Z. -4.314

54. Given $U = \{a,b,c,d,e, f,g,h,1,j,k,1\}$ as the universal set and its subsets as $P = \{a, b, f, g, j, l\}$, $Q = \{b,c,h,j\}$ and $R = \{a,d,e,f\}$. $P \cap R$ is
W. $\{l,k,i\}$ X. $\{a,f\}$
Y. $\{b,f,g\}$ Z. $\{d,l\}$

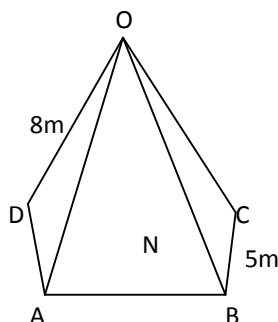
55. Given that $\tan\theta = 5/12$ then the actual value of $\sin^2\theta \pm \cos^2\theta$ is

56. The product of 214 and 23 in base 5 is

- W. 4922 X. 237
Y. 428 Z. 11032

57. The sum of 28 terms of the A.P. 3,10,17 _____

- W. 1427 X. 3816
Y. 4234 Z. 2730



58. The figure above represents a pyramid with a square base ABCD of side 5m. $OA = OB = OC = OD = 8\text{cm}$. What is the height ON of the pyramid?

- W. 5.37m X. 8.14m
Y. 6.35m Z. 7.18m

59. A cone of height 6cm and base radius 7cm is placed on top of the cross section of a cylinder of height 10cm and of the same radius with the cone. Find the volume of the composite figure so formed.

- W. 240cm^3 X. 1240cm^3
Y. 1624cm^3 Z. 1848cm^3

60. One interior angle of a regular polygon 170° . How many side has the polygon?

- W. 24 X. 43
Y. 18 Z. 36

61. If the kinetic energy of a system is conserved after a collision and the particles moves away with different velocities, then the collision is ____

62. Which of these methods can be used to determine the surface tension of a liquid? W. Vector method
X. Viscosity method
Y. Capillary method Z. Tension method

63. A pulley system has three ropes supporting a load of 35kgi 150 N is needed lift the load efficiency of the pulley ($g = 9.8\text{m/s}^2$) W. Hyper elastic Y. Perfectly elastic
X. Inelastic Z. Partly elastic

64. In relation to the movement of free electrodes in an electric current, which of the following is odd?

- W. Silicon X. Silver
Y. Copper Z. Aluminium

65. If a magnet is broken in the middle in an attempt to separate the poles. One finds two new magnets, each having pole(s)

- W. One X. Two
Y. Three Z. Four

66. The temperature at which the water present in the air saturates the air to condense is known as _____

- W. Dew point X. Melting
Y. Boiling Point Z. Freezing point

67. When a girl moves towards a plane mirror at a speed of 4.0m/s , the distance between the girl and her image reduces at a speed of _____

- W. 2m/s X. 4m/s
Y. 8m/s Z. 12m/s

68. A quantity which in direction to magnitude requires the specification to its direction to completely specify it is referred to as _____

- W. Derived quantity
X. Fundamental quantity
Y. Vector quantity
Z. Scalar quantity

69. A car running at 30m/s accelerate at 2m/s^2 for 20 seconds. What will be the final velocity of the car?

- W. 30m/s X. 50m/s
Y. 70m/s Z. 90m/s

70. At a height of 39,000m, an air force bomber dropped a bomb while maintaining a constant velocity of 1200km/h towards a point directly above its target. At what time will the bomb hit the target?

- W. 48.7 seconds X. 45.9 seconds
Y. 88.3seconds Z. 101.6 seconds

71. A machine gun with a mass of 50kg fire a 0.5kg bullet at a speed of 100m/s . The recoil speed of the machine gun is

- W. 0.5m/s X. 1.0m/s
Y. 1.5m/s Z. 2.0m/s

72. Germanium material can be classified as _____

- W. Conductor X. Super Conductor
Y. Semi Conductor Z. Insulator

73. The capacitance of a capacitor is said to _____ When the charge 'q' on it is doubled.

- W. Increased X. Decreased
Y. Remain the same Z. Tripled
74. A mixture of 0.20 mole of Ar, 0.20 of Ne, and 0.30 mole of He exerts a total pressure of 2.1 atmosphere. What is the partial pressure in the mixture?
W. 0.70 atmosphere X. 0.80 atmosphere
Y. 0.90 atmosphere Z. 0.65 atmosphere
75. A certain compound, on analysis gave 2.02g of Carbon, 0.35g of Hydrogen, and 2.68g of Oxygen. If the molecular mass is 90, calculate the molecular formula.
W. CH_2O X. $\text{C}_2\text{H}_4\text{O}$
Y. $\text{C}_3\text{H}_6\text{O}$ Z. $\text{C}_4\text{H}_8\text{O}_4$
76. An effect of thermal pollution in water bodies is that the
W. Volume of chemical waste increases
X. Level of oxides of nitrogen increase
Y. Level of oxygen reduces Z. Volume of water reduces.
77. Which of the following compounds is soapless detergent.
W. $\text{C}_{12}\text{H}_{25}\text{OSO}_3\text{Na}$ X. $\text{C}_{17}\text{H}_{25}\text{COONa}$
Y. $\text{C}_6\text{H}_5\text{OH}$ Z. $\text{C}_4\text{H}_9\text{OOC}_2\text{H}_5$
78. How many atoms are present in 6.0g of magnesium
W. 1.20×10^{22} X. 2.41×10^{22} Y. 1.51×10^{23}
Z. 3.02×10^{23} [Mg = 24, Na = 6.02 $\times 10^{23} \text{ mol}^{-1}$]
79. Which of the following pollutants is associated with brain damage?
W. Carbon (ii) Oxide
X. Radioactive fallout
Y. Biodegradable waste
Z. Sulphur (iv) Oxide
80. The maximum number electrons in the L shell of an atom is
W. 2 X. 8
Y. 18 Z. 32
81. Calculate the volume of $0.5 \text{ mol dm}^{-3} \text{ H}_2\text{SO}_4$ that is neutralized by 25 cm^3 of $0.1 \text{ mol dm}^{-3} \text{ NaOH}$
W. 5.0 cm^3 X. 2.5 cm^3
Y. 0.4 cm^3 Z. 0.1 cm^3
82. Which one of these is not found in an atom?
W. Electrons X. Molecules
Y. Neutron Z. Proton
83. The number of isomers that can be obtained from C_4H_{10} is
W. 4 X. 1 Y. 2 Z. 3
84. Which of the 3d, 4s, 4p, 4d has the lowest energy?
W. 3d X. 4s Y. 4p Z. 4d
85. Carbon is often deposited in the exhaust pipe of cars because of the
W. Dehydrogenation of petrol
X. Incomplete combustion of petrol
Y. Presence of additives
Z. Presence of carbon in petrol
86. Plaster of paris (P.O.P) used for making cast in hospitals is
W. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ X. $\text{CaSO}_4 \cdot \text{H}_2\text{O}$
Y. $\text{CaSO}_4 \cdot 3\text{H}_2\text{O}$ Z. $\text{CaSO}_4 \cdot 4\text{H}_2\text{O}$
87. Which of the following animals is cold blooded?
W. Cat X. Lizard
Y. Whale Z. Bird
88. Plants which can survive in place where water supply is limited are called:
W. Mesophytes X. Hydrophytes
Y. Heliophytes Z. Xerophytes
89. All but one of the following is a component of plant cells:
W. Cell membrane X. Hydrophytes
Y. Vacuole Z. Ribosome
90. The study of algae is called:
W. Bacteriology X. Algology
Y. Mycology Z. Psychology
91. The tilapia fish uses ____ fin for balancing:
W. Anal X. Ventral
Y. Caudal Z. Dorsal
92. The division of the fungi in which the sexual phase is unknown is called:
W. Deuteromycotina
X. Zygomycotina
Y. Ascomycotina
Z. Basidiomycotina

93. The digestive enzyme that coagulates proteins in milk is _____
 W. Ptyalin X. Pepsin
 Y. Tripsin Z. Renin
94. During reproduction in Amoeba the first structure to divide is:
 W. Cytoplasm X. Nucleus
 Y. Vacuole Z. Cell Membrane
95. Euglena is classified as a plant due to the presence of:
 W. Flagellum X. Chloroplast
 Y. Gullet Z. Cell membrane
96. The end products of the digestion of fats and oil are:
 W. Fatty acids and glycerol
 X. Protein and oil
 Y. Amino-acids and fats
 Z. Glucose and fats
97. One of the following is not a factor required for seed germination:
 W. Light X. Food
 Y. Air Z. Viable seed
98. In somatic cell appears in:
 W. Haploid X. Diploid
 Y. Triploid Z. Polyploid
99. A ripened or mature ovary is also known as:
 W. Fruit X. Flower
 Y. Embryo Z. Seed
100. Cells are differentiated into different tissues in _____ regions:
 W. Meristematic X. Elongation
 Y. Differentiation Z. Maturation

SOLUTION

1. Z 2. W 3. X 4. W 5. W 6. Y 7. X
 8. W 9. Z 10. Y 11. X 12. Z 13. W 14. W
 15. X 16. X 17. Z 18. X 19. X 20. W 21. W
 22. Y 23. W 24. Y 25. W 26. Z 27. Z 28. W
 29. Z 30. Y 31. X 32. Z 33. X 34. W 35. Y
 36. Y 37. W 38. X 39. Y 40. Y 41. X 42. Z
 43. Z 44. W 45. Y 46. W 47. X 48. Z 49. X
 50. Y 51. Z 52. Y 53. Y 54. X 55. Y 56. Z
 57. Z 58. 59. Z 60. Z 61. Y 62. Y 63. X
 64. W 65. X 66. W 67. X 68. Y 69. Y 70. Y

71. X 72. Y 73. W 74. 75. Y 76. Y 77. W
 78. Y 79. X 80. X 81. X 82. X 83. Y 84. Z
 85. X 86. W 87. X 88. Z 89. 90. Z 91. Z
 92. W 93. Z 94. X 95. X 96. W 97. X 98. X
 99. W 100. W

POST-JAMB SCREENING TEST REVISION QUESTIONS AND ANSWERS 2017/2018 SESSION DELSU

ENGLISH

From 1 to 6 choose the option that best complete each of the following sentences:

- The _____ of the food filled the whole compound
 W. Fragrance Y. Aroma
 X. Scent Z. Perfume
- He was caught distributing political ____
 W. Brochures Y. Bulletins
 X. Circulars Z. Pamphlets
- She _____ her spoon in the soup and tasted it
 W. Inserted Y. Buried
 X. Sank Z. Dipped
- When rescued from the sea, he was ____ with cold.
- He was so angry he couldn't talk ____
 W. Efficiently X. Capably
 Y. Proficiently Z. Coherently
- He delighted them with his _____ stories
 W. Interminable X. Inexhaustible
 Y. Immeasurable Z. Illimitable

From questions 7 to 8 choose the option nearest in meaning to the underlined

- You should learn to live within your meagre means
 W. Plenty Y. Inadequate
 X. Surplus Z. Immense

8. Soyinka's plays will not be ephemeral W. Multi-coloured Y. Built-in
X. Short-lived Z. Long-lived

From 9 to 11 choose the option that has the same vowel sound as the one underlined

9. Sea W. Sit Y. Piece
X. Realm Z. Rich
10. Set W. Remain Y. Net
X. Seize Z. Pretty
11. Pass
W. Plait Y. Rash
X. Marry Z. Part

Read passage 1 carefully and answer the questions 12-16

Advertisements are part of our everyday lives. We meet them in newspapers, on the radio and television, in the street and even in buses and trains. We are so accustomed to them that it would be very strange to wake up one morning and find that they had all disappeared.

Yet advertising by manufacturers only occurs on a large scale in democratic countries, where free enterprise leads to competition and competition to advertisement. In certain totalitarian countries where the state controls production, there is a cut-throat competition but equally there are very few brand names and only a meagre choice of articles. In these circumstances, most of the brand names are naturally well-known. If there is, for instance, only one brand of toothpaste, it is pointless to advertise it.

On the other hand, people often feel that competing manufacturers spend so much money on advertising their various brands that they have to increase the price of their articles considerably in order to cover the advertising costs. Although a lot of money is spent on advertising, this only raises the selling price of each mass-produced article by an insignificant amount. And from the manufacturer's point of view, such a small price reduction would be a poor selling device. It might attract a few purchasers who would not otherwise buy the article, but their number would be nothing in comparison with those who are attracted by advertisements.

Even though advertisements are the result of competition they do have a useful function: that of giving information. Of course, people often complain that - with all their exaggerations, half truths and distorted facts - advertisement misinform rather than inform. But do any of us really believe what we are told in an advertisement?

12. Which of the following is true according to the passage?
W. Advertisement is very costly
X. Advertisement leads to high cost of articles
Y. Advertisement is not necessary
Z. Advertisement is a common phenomenon
13. Advertisement is not common in ____
W. Totalitarian setting
X. Democratic societies
Y. Indigenous settings
Z. Demographic experiments
14. Advertisement is basically an off-shoot of ____
W. Competition Y. Ignorance
X. Anxiety Z. Inconsistency
15. Advertisement informs and mis-informs. Therefore it is
W. Double Y. Ambiguous
X. Careless Z. Sadistic
16. The last statement in the passage is an example of ____
W. Rhetorical question X. Hyperbole
Y. Oxymoron Z. Personification

From 17 to 19 choose the option that is opposite in meaning to the underlined and which can also fill the gap,

17. We were defeated but our junior team was ____
W. Vicarious Y. Pummelled
X. Protected Z. Victorious
18. Mr. Omoni made a lot of today unlike the ____ of yesterday
W. Purchase Y. Loss
X. Share Z. Gross
19. They usually labours when talking, but Amaka she speaks ____
W. 'Fluently Y. Fearfully
X. Painfully Z. Forcefully

From 20 to 22 choose the option that best completes the gap.

20. He came ____ the room and sat down
W. Over Y. Around
X. Into Z. Off
21. John has been ill ____ last Friday.
W. Before Y. Till

22. X. After Z. Since
Will you please ____ me your book?
W. Lend Y. Borrow
X. Supply Z. Excuse

In each of questions 23 to 25 select the option that best explains the information conveyed in the sentence

23. She is very right fingered W. She is skilful with her hands
X. She is apt to take things
Y. She has a gentle touch
Z. She is careless
24. The teacher turned a blind eye to the lateness of Mary W. He did not notice it
X. He disapproved of it
Y. He did not bother about it
Z. He was unable to detect it
25. The dispute has now come to a head. W. It has reached a gigantic proportion
X. It has reached a crucial stage
Y. It has reached the head of the manager
Z. It has got to the head office

APTITUDE TEST

Identify the Odd one in the options for each number

26. W. Ibadan Y. Enugu X. Jos Z. Kaduna
27. W. Rotimi Amaechi X. Kayode Fayemi
Y. Idris Wada Z. Sanusi Lamido

For each number below 28-30 identify the anagram to the given word

28. STAPLE W. Pin Y. Apple
X. Please Z. Plates
29. CANOE W. Ocean Y. Boat
X. Kano Z. Cane
30. SLAIN W. Saint Y. Pain
X. Nails Z. Kill
31. The 2014 FIFA world cup will take place in
W. Spain Y. South Africa
X. Japan Z. Brazil
32. The Governor of Kano State is
W. Ibrahim Shema

- X. Rabi Kwankwanso
Y. Jonah Jang
Z. Babangida Aliyu

33. The Headmasters of the African Union is located in
W. Aba Y. Adis Ababa
X. Cairo Z. Johannesburg

Use the logic below to find the missing numbers in questions 34 to 35

Logic: $6 \rightarrow 16$ $2, 1 \rightarrow 20$, and $\frac{1}{2} \rightarrow 12$ $5\frac{1}{2}$

34. $4 \rightarrow [] 6$
W. 20 Y. 24
X. 10 Z. 18
35. $3\frac{1}{2} \rightarrow [] 7$
W. 21 Y. $4\frac{1}{2}$
X. $10\frac{1}{2}$ Z. 19

Indicate the odd one from the options No 36-38

36. W. Doctor Y. Engineer X. Nurse Z. Pharmacist
37. W. Mosque Y. Minaret
X. Church Z. Koran
38. W. Gowon Y. Shagari
X. Buhari Z. Babangida

Choose the word that best completes the following phrases

39. Dog is to bark as pig is to ____
W. Hum Y. Rumble
X. Grunt Z. Squeak
40. Duck is to ____ as bee is to hum
W. Chatter Y. Blare
X. Crow Z. Quack
41. Cat is to kitten as horse is to ____
W. Cub Y. Colt
X. Kid Z. Ass
42. Abuja is to Nigeria as ____ is to Uganda
W. Lusaka Y. Kampala
X. Harare Z. East Africa

NOTE 4A 5 15, 6A 7 = 35 and 9A 1 = 8 use this logic to find the missing number in questions 43 to 46

43. $9A8 = []$
 W. 63 Y. 64
 X. 72 Z. 70
44. $4A(3A2) = []$ W 12 Y 24
 W. 12 Y. 24
 X. 20 Z. 18
45. $3A[] 1 = 10$
 W. $4\frac{1}{2}$ Y. 3
 X. 4 Z. 5
46. $(5A7) A2 = []$
 W. 54 Y. 56
 X. 68 Z. 48

Use the following logic to questions 47-50

Logic 3 N 4 = 14, 2 N 3 = 106 N 2 = 16

47. $17N[] = 40$
 W. 23 Y. 2
 X. 3 Z. 6
48. $[]N2 = 40$
 W. 18 Y. 14
 X. 20 Z. 28
49. $1N[] = 6$
 W. 4 Y. 3
 X. 1 Z. 2
50. $9N4 = []$
 W. 26 Y. 18
 X. 36 Z. 40

GENERAL PAPER

51. The quadratic equation whose roots are and 3 is.
 W. $x^2 + x - 12 = 0$ X. $x^2 + 7x + 12 = 0$
 Y. $x^2 - 2x + 12 = 0$ Z. $x^2 + 5x - 12 = 0$
52. If the opposite parallel sides of a trapezium is 8m and 1 2m, find the area if its height is 4m.
 W. $28m^2$ Y. $54m^2$
 X. $36m^2$ Z. $40m^2$
53. Two numbers are such that their sum is 18. Suppose that one is 8 greater than the other, find the numbers.
 W. 13 and 5 Y. -13 and 5

X. 10 and 8 Z. -10 and 8

54. y is inversely proportional to x and y = 4 when x = $\frac{1}{2}$. Find x when y = 10
 W. $\frac{1}{10}$ Y. 2
 X. $\frac{1}{5}$ Z. 10
55. Two angles in a triangle are 87° and 33° . The remaining angle is
 W. 72° Y. 60°
 X. 90° Z. 120°
56. A line touching the circumference from outside a circle is called
 W. Secant Y. Radius
 X. Chord Z. Tangent
57. If $132 = 42$ where x is a number base and the right hand side is in denary. Find x
 W. 6 Y. 8
 X. 5 Z. 4
58. What are the possible values of x given that $\log_2(x^2 + 5x + 14) = 3$? W. -4, 3 Y. -1, -5
 X. -3, -2 Z. -5, -7
59. A trader sold a bicycle at N10,000 with 25% profit. His cost price is
 W. N 2,000 Y. N 6,000
 X. N 8,000 Z. 14,000
60. Three pencils and two ruler cost N13 while two pencils and one ruler cost N8. How much is the cost of one pencil and one ruler respectively?
 W. N5, N3 Y. N2, N7
 X. N4, N6 Z. N3, N2
61. The geometric mean of 2 and 98 is
 W. 8 Y. 18
 X. 24 Z. 14
62. 15% of a certain number is 6. What is the number?
 W. 20 Y. 60
 X. 40 Z. 80
63. Convert 257 in base ten to base 12
 W. 160 Y. 195
 X. 221 Z. 215
64. How many seconds are in 5 hours? W. 72240secs Y. 18000secs

- X. 16060secs Z. 19080secs
- W. 1.5sec Y. 16.7sec
X. 6.0sec Z. 150sec
65. What is the Celsius temperature corresponding to 100 degrees Kelvin?
W. -173°C Y. 100°C
X. 0°C Z. 173°C
66. The temperature of hot and cold end of a 20cm long rod in thermal steady state are at 100°C and 200°C respectively. What is the temperature of the centre of the rod?
W. 60°C Y. 40°C
X. 50°C Z. 30°C
67. The Calorie is defined as the quantity of heat required to raise the temperature of
W. 1g of water by 1°C
X. 1g of water by 100°C
Y. 1g of water by 4°C
Z. 1g of water by -273K
68. A 25 watt, 220 volt bulb and a 100 watt, 220 volt bulb are connected in series across a 220 volt line. Which electric bulb will glow more brightly.
W. 25 watt bulb
X. 100 watt bulb
Y. 130th glow at the same rate
Z. Both with same brightness
69. Used normally, a 150W, 120V light bulb requires how many amperes of current?
W. 1.25A. Y. 3.00A X. 2.50A. Z. 0.80A.
70. Which one of the following laws best describes the operation of an electric generator?
W. Ampere's law Y. Ohm's law
X. Gauss' law Z. Faraday's law
71. A particle is oscillating according to the equation $x = 7\cos 0.5\pi t$, where t is in second. The point moves from the position of displacement to maximum displacement in time $t = \underline{\hspace{1cm}}$
W. 4.0sec Y. 1.0sec
X. 2.0sec Z. 0.5sec
72. A ball leaves a girls hand with an upward velocity of 6 m/s. What is the maximum height of the ball above the girl's hand?
W. 1.8m Y. 12.0m
X. 6.0m Z. 18.0m
73. The time needed for a net force of 10 N to change the velocity of a 5 kg mass by 3 m/s is
74. A converging thin lens has a focal length of 27cm. An object is placed 9cm from the lens. What is the image distance from the lens.
W. 13.5cm Y. 3.0cm
X. 18.0cm Z. 36.0cm
75. When the image formed by a convex mirror is compared to its object, the image is usually
W. Erect and smaller
X. Inverted and imaginary
Y. Real and inverted
Z. Real and erect
76. By which of the following can a real image be produced?
W. Concave mirror X. Plane mirror
Y. Diverging lens Z. Convex mirror
77. If concentrated H_2SO_4 is added to sugar and warmed gently, the sugar changes from white to brown and finally to a black mass of carbon. In this reaction, concentrated H_2SO_4 is acting as:
W. A drying agent X. An oxidizing agent
Y. A dehydrating agent Z. A reducing agent
78. Which of the following can have an isomer?
W. C_2H_4 Y. C_3H_4
X. C_4H_{10} Z. C_6H_6
79. The final products of the reaction between methane and chlorine in the presence of ultraviolet light are hydrogen chloride and:
W. Chloromethane
X. Tetrachloromethane
Y. Trichloromethane
Z. Dichloromethane
80. In the electrolysis of brine, the anode is
W. Platinum Y. Zinc
X. Copper Z. Carbon
81. In the electrolytic extraction of aluminium, the function of molten cryolite is to
W. Precipitate aluminium hydroxide
X. Lower the melting point of aluminium oxide
Y. Act as raw material
Z. Act as a solvent
82. 0.05 mol dm^{-3}HCl is neutralised by

- 25cm³NaOH. If the solvent volume of the acid used is 32.00cm³. What is the concentration of the base?
- W. 0.016mol dm⁻³ X. 10.032mol dm⁻³
Y. 0.039mol dm⁻³ Z. 0.064mol dm⁻³
83. The periodic classification of the elements is an arrangement of the elements in order of their.
W. Atomic mass X. Atomic number
Y. Atomic volume Z. Molecular weight
84. Helium atoms are chemically unreactive because:
W. There are no electrons around the nucleus
X. The number of protons equal the number of electrons
Y. The atoms contain only protons
Z. The outer electron shell is completely
85. How many grams of hydrogen gas will be liberated when 6g of magnesium ribbon dissolves in 500cm³ of 6MHCl?
W. 1.2g Y. 0.5g
X. 0.7g Z. 0.3g
86. A sample of orange juice is found to have a pH of 3.80. what is the concentration of the hydroxide ion in the juice?
W. 1.6×10^{-4} X. 6.3×10^{-11}
Y. 6.3×10^{-4}
87. $\text{CaO}_{(s)} + \text{H}_2\text{O}(\text{ }) \rightarrow \text{Ca}(\text{OH})_{2(s)} \Delta H = -65\text{kJ}$. The process represented by the above equation is known as
W. Dissolution Y. Liming
X. Slaking Z. Mortaring
88. The modern process of manufacturing steel from iron is by
W. Treatment with acid
X. Oxidation
Y. Blast reduction
Z. Treatment with alkalis
89. Ovoviviparity is the type of fertilization exhibited by the organism labelled
W. I Y. III
X. II Z. IV
90. Thunderstorm can be beneficial to plants because
W. Kills the pest that affect crops
X. Destroy some of the major crops
Y. Makes rain water available to soil
Z. Add nitrates to the soil
91. The organic molecules present in the nucleus is/are
W. DNA X. Protein, RNA and DNA
Y. RNA and ATP Z. RNA
92. A characteristic of the phylum Coelenterata is that
W. Most of them are marine
X. They possess a gut with a single opening.
Y. They possess numerous pores on their body
Z. They are bilaterally symmetrical
93. The smallest living organisms which share the characteristics of both living and non-living matter are
W. Bacteria Y. Viruses
X. Fungi Z. Protozoa
94. 2.8g soil sample was heated to a constant weight of 24g. When further heated to red hot and cooled, it weighed 18g. What is the percentage of humus in the soil.
W. 22.2 Y. 75.0
X. 55.6 Z. 25.0
95. Mr. Ochuko usually works in a swamp during which he neither eat food nor drink any water. Which of these diseases Mr. Ochuko cannot contact.
W. Cholera X. Sleeping sickness
Y. Bilharziasis Z. River blindness
96. One of the following statements is not true of viruses
W. They are microorganisms
X. They are smaller than bacteria
Y. They cause polio and small pox
Z. They can be seen by ordinary light microscope
97. A gene which expresses itself only in the homozygous condition is
W. A Mutant Y. Lethal
X. Dominant Z. Recessive
98. Which of the following represents the phenotypic ratio of the above genetic cross.
W. 1:2:1 Y. 3:1
X. 4:1 Z. 2:2

99. The surface of an alveolus in a mammal is well supplied with tiny blood vessels known as
W. Arterioles Y. Arteries
X. Venules Z. Capillaneo
100. The nitrogeneous substance that is excreted by birds in order to conserve water is
W. Ammonia Y. Urea Acid
X. Uric Acid Z. Nitric Acid

SOLUTION

1. Y 2. Z 3. Z 4. W 5. Z 6. X 7. Y
8. X 9. Y 10. Y 11. Z 12. Z 13. W 14. W
15. Y 16. W 17. Z 18. Y 19. W 20. X 21. Z
22. W 23. X 24. Y 25. X 26. X 27. Z 28. Z
29. W 30. X 31. Z 32. X 33. Y 34. W 35. W
36. Y 37. X 38. Y 39. X 40. Z 41. Y 42. Y
43. Y 44. W 45. Z 46. W 47. X 48. W 49. Z
50. W 51. W 52. Z 53. W 54. X 55. Y 56. Z
57. X 58. X 59. X 60. Z 61. Z 62. X 63. Y
64. Y 65. W 66. Y 67. W 68. X 69. W 70. Z
71. W 72. W 73. W 74. W 75. W 76. W 77. Y
78. X 79. X 80. Z 81. X 82. Z 83. X 84. Z 85. Z 86. X 87. X
88. X 89. 90. Z 91. X
92. X 93. Y 94. Z 95. Z 96. Z 97. X 98.
99. X 100. Y

POST-JAMB SCREENING TEST REVISION QUESTION AND ANSWERS 2016/2017 SESSION DELSU

ENGLISH LANGUAGE

Instruction: Read the passage below and answer the questions that follow.

The endeavour to maintain proper standards of fairness in Journalism must be pursued. It is fatally easy for the Journalist to deviate from the straight path. There is his natural desire to 'make a story' and insidious temptation to twist facts to square with his paper's policy. Both indefensible as the framing of misleading headlines for the sake of effect. The conscientious Journalist must check any tendency to bias, and guard against the dangers inherent in personal antipathies of friendships, and in traditional opposition between rival schools of thought. When a political opponent, whose stupidity habitually provokes attack, makes an effective speech, honesty requires that he should be given credit for it.

1. The duty of the Journalist to the public entails
A. Telling the truth

- B. Trying to make a story
C. Keeping quiet when necessary
D. Defending the weak

2. A dutiful Journalist must
A. Be able to make an effective speech
B. Have political acumen
C. Be able to recognise merit wherever found
D. Read widely

3. Insidious as used in the passage means
A. Wild B. Wicked
C. Unusual D. Subtle

4. When may a journalist have to act against his own inclination?
A. After he has been given a bribe
B. When his enemies make a mistake
C. When he become weak and decrepit
D. When his friends acts stupidly

5. A Journalist would be able to uphold the tenants of his profession if:
A. He tries to protect his friends B. He resists temptation to bias.
C. He seeks publicity
D. He tries to achieve publicity

Instruction: Choose the option that conveys the meaning of the underlined portion in each of the following sentences

6. The result of Dr. Ese's experiment represents a breakthrough science. A. Catastrophe
C. A breaking point
C. An outstanding success
D. An end to such experiments
7. When the man was caught by the police, he presented a bold front
A. He faced the situation with apparent boldness
B. He attacked the policeman boldly
C. He bravely attempted to bribe the policeman
D. He frowned at them in a deficient manner
8. Publishers say that newsprint has become more expensive
A. Newspapers
B. Publishing news
C. Paper for printing
D. Printer's salaries and wages.

9. His attempts were nipped in the bud A. Frustrated
B. Unsuccessful
C. Cut short D. Disqualified
10. The task was herculean A. Irregular
B. Related to herculis
C. Demanding D. Stimulating
11. The sudden death of the king put paid to the ambition of the minister
A. Rewarded B. Encouraged
C. Benefited D. Terminated
12. Edafe was a mindless criminal
A. Heartless B. Mindful
C. Strong D. Memorable

Instruction: Complete each of the following Sentences by choosing the option that best suitably fills the space

13. Why do you worry about such ____ matters?
A. Non-significant B. Unsignlficant
C. Insignificant D. Significant
14. The Pastor said 'lay ____ your hearts to God'
A. Bear B. Bare
C. Bar D. Barred
16. What a relief that the strike ____ when it did
A. Ended B. Would ended
C. Has ended D. Is ending
17. I saw you walking in that direction, but I did not know exactly_____
A. Where you are going
B. Where are-you going
C. Where you were going
D. To where were you going
18. Do you know if the new teacher _____ yet
A. Has arrive B. Will be arriving
C. Has arrived D. Has been arriving

Instruction: Choose the word which is opposite in meaning to the underlined word in each sentence

19. Unless we see the evil clearly, contending against them is like fighting the air
A. Avoiding B. Struggling with
C. Combating D. Repelling

20. Nigeria is gradually recession A. Destruction
B. Recovery
C. Discovery D. Birth
21. Since mortality is the lot of man, argued the preacher, we must all accept death with equanimity.
A. Anxiety B. Concern
C. Uncertainty D. Serenity
22. I was advised to stay away from cities that are ravaged by crime
A. Spoilt by B. Suffering from
C. Haven of D. Free from
23. Adama's father simply condoned her bad habits instead of punishing her appropriately
A. Opposed B. Blamed
C. Accepted D. Forbade
24. The debtors husband is liable for his wife's A. unanswerable B. Unquestionable
C. Responsible D. Accountable
25. In an English sentence, there is always a concord between the subject and the verb
A. Disagreement B. Harmony
C. Harmoniousness D. Agreement

APTITUDE

26. Which of the following numbers is prime?
A. 117 B. 147
C. 159 D. 149
27. A boy was asked to multiply a number by $\frac{7}{8}$ instead he divided the number by $\frac{7}{8}$ and got the result 15 more than the correct result The number was
A. 96 B. 56
C. 36 D. 65
28. The eighth term of 9,6,3,0 ... is
A. 8 B. -12
C. -14 D. 6
29. The expression $\frac{11}{5} + \log \frac{14}{3} - \log \frac{22}{15}$ is equal to
A. Log 2 B. Log 7
C. Log 3 D. Log 5
30. The difference between the place values of 7 and 3 in the number 527435 is

- A. 4070 B. 6790
C. 3690 D. 6970

Direction to question 31 to 35: The following questions consists of two words each of them have certain relationship to each other, followed by four lettered pairs of words. Select the lettered pair that has the same relationship as of the original pair of words.

31. Trees: Branches
A. River: Tributaries B. Continent: Island
C. Stream: Delta D. Ocean: Seas
32. Relax: work
A. Play: Cheat B. Lunch: Dinner
C. Smile: Laugh D. Fresh: Stale
33. Square: Cube
A. Line: Cylinder B. Triangle: Prism
C. Circle: Sphere D. Sphere: Earth
34. Carnivore: Herbivore
A. Animal: Bird B. Camel: Giraffe
C. Flesh: Plant D. Horse: lion
35. Accident: Mishap
A. Abrupt: Sudden B. Eminent: Notorious
C. Pacify: Provoke D. Dearth: Plenty
36. If train is called bus, bus is called tractor, tractor is called car, car is called scooter, scooter is called bicycle, bicycle called moped, which is used to plough a field?
A. Train B. Bus
C. Tractor D. Car
37. Pointing to a man on the stage, Rita said 'He is the brother of the daughter of the wife of my husband'. How is the man on the stage related to Rita?
A. Son B. Cousin
C. Husband D. Brother-in-law
38. Rahul told Anand "Yesterday defeated the only brother of the daughter of my grandmother"
A. Son B. Brother
C. Brother D. Father-in-law
39. A man is facing south. lie turns 135 degrees in the anticlockwise direction and then 180 degree in

clockwise direction. Which direction he is facing now?

- A. North-East B. North – West
C. South-East D. South-West

40. 36 vehicles are parked in the single row. After the first car there is one scooter after the second car there are scooters after the third car there are three scooters and so on work out the number of scooters in the second half of the
A. 10 B. 12
C. 15 D. None of these
41. A number greater than 3 but less than 8 also it is greater than 6 but less than 10. The numbers.
A. 5 B. 6
C. 6 D. 9
42. If the length and breath of rectangular plot are increased by 50% and 20% respectively, the new areas will be how many times than that of original area?
A. 4 B. 9/6
C. 11/5 D. 9/5
43. A mixture contains alcohol and water in the ratio of 4:3. If 7Ltrs of water is added to the mixture the ratio of alcohol and water becomes 3:4. Find the quantity of alcohol in the mixture.
A. 12 Ltrs. B. 8Ltrs C. 9 Ltrs.
D. 15 Ltrs.

Directions for questions 44 to 47: Each question has a part of the sentence or the entire sentence underlined, this is followed by four different ways of phrasing the sentence. Choose the alternative which is grammatically correct and concise.

44. More than 35 school children. Some of them who were as young as nine. Were killed when an explosion ripped through their primary school.
A. more than 35 school children, some as young as nine
B. More than 35 school children, some as young as nine
C. As Many as 35 school children, some as young as nine
D. Not less than 35 school children, some as young as nine
45. In view of the reasons that have been cited. I hereby recommended passing of the resolution.
A. In lieu of the reason given

- B. Because the reasons have been given
C. In view of the reasons given
D. In view of the reasons that have been cited
46. Mrs. Pooja, our class teacher and who is also a member of the school committee can speak excellent German.
A. Who is also a member of the school committee can speak excellent German
B. Who also is a member of the school committee can speak excellent German
C. Also a member of the school committee can speak in German
D. A member of the school committee can speak excellent German.
47. As Sohan was on probe he was determined to not fall foul of the law.
A. Was on probe he was determined to not fall foul of the law
B. Was on probe. He was determined not to fall foul of the law
C. Was in probe, he was eager to avoid the law
D. Was probe, he did not with to provoke the law
48. If the letters of the word EQUATION' be arranged at random, what is the probability that all vowels are together.
A. $\frac{1}{8}$ B. $\frac{1}{12}$
C. $\frac{1}{16}$ D. $\frac{1}{14}$
52. The liquid in a cone of radius r cm and height 12cm is poured into a cylinder of the same base radius. Calculate the height of the water in the cylinder
A. 8cm B. 14cm
C. 6cm D. 4cm
53. Solve $2^{2x} \times 5(2^x) + 4 = 0$ A. 3 or -5 B. 0 or 2
C. 1 or 4 D. -3 or 1
54. Factorise $125x^2 - 1$
A. $(5x + 1)(25x^2 - 1)$
B. $(25 + 1)(10x^2 + 5x + 1)$
C. $(5x - 1)(25x^2 + 5x + 1)$
D. $(50x - 1)(5x^3 \pm 3x^2 + x - 1)$
55. If y varies inversely as the cube root of x and $y = 6$ when $x = 64$ find y when $x = 125$
A. $\frac{55}{6}$ B. $\frac{33}{4}$ C. $\frac{44}{5}$ D. $\frac{66}{7}$
56. If $P\%$ of a number n equal k , what is 7% of n ?
A. $\frac{7k}{100p}$ B. $\frac{k}{200p}$
C. $\frac{k}{3p}$ D. $\frac{7k}{p}$
57. For what values of x is the expression $x^2 - 4x - \frac{1}{x^2} + x - 2$ undefined?
A. -2, 1 B. -3, 2
C. -1, 4 D. -5, 2

Direction for question 49 to 50: Answer the questions based on the following information. N is a natural number and it has only four distinct factor- 1, a , B and N itself (Assume that a is always less than b)

49. Integer a is a _____
A. Prime number
B. Odd but not prime number
C. Even number
D. Composite number
58. The bearing of A from B is $N10^\circ W$. Find the bearing of B from
A. 170° B. 180°
C. 165° D. 120°
59. If the mean of 5, x , 8, 14, 10, 16 is 10, find the median
A. 12 B. 9
C. 11 D. 13
60. The first term of an AP is 3. If the sum of the first 8 terms is 80, then the common difference is?
A. 2 B. 7
C. 9 D. 4

GENERAL PAPER

51. There are 40 identical balls in a box. If the balls are either red or white, and the probability of picking a red ball is $\frac{1}{4}$ then the total number of white balls is
A. 30 B. 20
C. 10 D. 40
61. The length of a rectangle is 4cm longer than its breadth. If the area is 60cm^2 find its length.
A. 14cm B. 10cm
C. 12cm D. 8cm
62. Find x if $\log 9 = \frac{1}{2}$
A. -8cm B. 5
C. 5 D. 3

63. A rectangular carpet measures 16m by 12m and it has stained borders of width 2m surrounding it. What is the area of the boarder
A. 360m B. 128m²
C. 221m D. 400m²
64. An Ogive is perfectly called
A. Frequency polygon
B. Cumulative frequency curve
C. Frequency distribution
D. Curve
65. An object is dropped from rest from a certain height. After a distance d, the object's kinetic energy is proportional to which of the following?
A. $1/d^2$ B. $1/d$
C. d D. d^2
66. A particle of mass 100 g is thrown vertically upwards with a speed of 5m/s. The work done by the force of gravity during the time the particle goes up is.
A. -0.5J B. -1.25J
C. 0.5J D. 1.25J
67. What is the length of a simple pendulum that has a period of 2.8 seconds ($g = 9.8 \text{ m/s}^2$)
A. 60cm B. 91cm
C. 136cm D. 195cm
68. A wooden object whose density is 0.06 g/cm^3 was dropped into water contained in a tall beaker. How much of the wooden object is in the water?
A. small fraction B. One-sixth
C. One-quarter D. Two third
69. The quantity of heat in a body is depended on which of the following. A. Temperature
B. Specific Heat Capacity
C. Mass
D. All of these
70. When heat is applied to a brass metal, it expands because its molecules____
A. Vibrate with smaller amplitude
B. Vibrate with greater amplitude
C. Increase their speed of random motion
D. Decrease their speed of random motion
71. 80 cm^3 of a gas is measured at 8°C and 740 mmHg. What is the volume of the gas at standard temperature and pressure. A. 75.7 cm^3 B. 79.3 cm^3
C. 84.1 cm^3 D. 87.5 cm^3
72. For a sound speed of 350m/s, what would be the wavelength of a 5000 Hz sound?
A. 2cm B. 3.5cm
C. 5cm D. 7cm
73. Which of the following attributes of a musical note helps the ear to differentiate between different notes on the basis of their source.
A. Pitch B. Intensity
C. Quality D. All of these
74. Which of the following instruments could be used in finding the angle of elevation of the sun?
A. Periscope B. Telescope
C. Kalcidoscope D. Sextant
75. In a transformer, the number of turns of primary coil and secondary coil are 5 and 4 respectively. If 220 volts is applied on the primary coil, then the ratio of primary current to the secondary current will be
A. 4:5 B. 5:4
C. 5:9 D. 9:5
76. The plate of parallel plate capacitor of area 3 m^2 are kept 8 mm apart in a vacuum. What is the capacitance of the capacitor ($\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$)?
A. $3.0 \times 10^{-19} \text{ F}$ B. $4.0 \times 10^{-19} \text{ F}$
C. $5.0 \times 10^{-19} \text{ F}$ D. $5.5 \times 10^{-19} \text{ F}$
77. The efficiency of a solid catalyst to increase reaction rates depends on its
A. Solubility B. Surface area
C. Concentration D. Quantity
78. The volume of 2.5 moles of oxygen at S.T.P is (Molar volume of gas at STP. = 22.4 dm^3)
A. 50.0 dm^3 B. 84.0 dm^3
C. 28.0 dm^3 D. 56.0 dm^3
79. Which of the following represents an aromatic compound?
A. $\text{C}_6\text{H}_5\text{OH}$ B. C_6H_{12}
C. $\text{C}_6\text{H}_9\text{OH}$ D. C_6H_8
80. A compound contains 31.91% potassium, 28.93% Chlorine and the rest oxygen. What is the chemical formula of the compound?

(K=39, CL=35.5, O=16).

- A. KClO_4 B. KClO
C. KClO_2 D. KClO_3

81. By 25cm^3 of 0.2mol dm^{-3} solution of Na_2CO_3 requires 20cm^3 of a solution of HCl for neutralization. The concentration of the HCl solution is
A. 0.5mol dm^{-3} B. 0.6mol dm^{-3}
C. 0.2mol dm^{-3} D. 0.4mol dm^{-3}

82. A gas X diffuses twice as fast as gas Y under the same conditions. If the relative molecular mass of X is 28, calculate the relative molecular mass of Y.
A. 112 B. 120
C. 56 D. 14

83. Which of the following ion is acidic?
A. K^+ B. NO_3^-
C. S_2^{2-} D. H_3O^+

84. In the electrolysis of a concentrated solution of sodium chloride using inert electrode, which of the following ions are discharged at the cathode and the anode respectively?
A. Na^+ and Cl^- B. Na^+ and OH^-
C. H^+ and OH^- D. H^+ and Cl^-

85. A given volume of methane diffuses in 20s. How long will it take the same volume of sulphur (IV) oxide to diffuse under the same conditions? [$\text{C}=12, \text{H}=1, \text{S}=32, \text{O}=16$]
A. 5s B. 20s
C. 40s D. 60s

86. What volume of 0.5mol dm^{-3} H_2SO_4 will exactly neutralize 20cm^3 of 0.1mol dm^{-3} NaOH solution?
A. 2.0cm^3 B. 6.8cm^3
C. 5.0cm^3 D. 8.3cm^3

87. What is the correct IUPAC name for NO_2^- ?
A. Dioxonitrate (IV) ion
B. Trioxonitrate (IV) ion
C. Trioxonitrate (III) ion
D. Dioxonitrate (III) ion.

88. The pressure of 100cm^3 of oxygen at 35°C , is 750mmHg . What will be the volume of the gas if the pressure is reduced to 100mmHg without changing the temperature.
A. 650cm^3 B. 850cm^3

C. 580cm^3 D. 750cm^3

89. Which of the following is not an outcome of ecological succession:

- A. There is progressive change in species structure, organic matter energy flow
B. The climax stage consists of a comparatively long lasting species composition.
C. There are fewer dominant species in the late stages of succession
D. There is no dynamic equilibrium between the community and habitat in the climax.

90. The statement that the environment determines which organism survives to reproduce:

- A. Explains the theory of acquired characters
B. Support Jean Lamarck's theory of use and disuse
C. Supports the idea of survival of the fittest
D. Implies that organisms modify their environment in order to survive

91. The young chick is formed from

- A. 3 B. 3 and 4
C. 1, 3 and 5 D. 2

92. Which parts provide food for the developing chick?

- A. 2 and 3 B. 4 and 5
C. 3 and 4 D. 1 and 2

93. Which of the following sequence represent the process of blood clotting? 1. Fibrin forms a network of threads 2. Red blood cells are caught and a clot is formed. 3. Fibrinogen in plasma changes into soluble fibrin 4. Blood is exposed to air.

- A. 4, 3, 2, 1 B. 4, 3, 1, 2
C. 3, 1, 4, 2 D. 1, 2, 3, 4

94. Groundnut is not really a 'nut' in the biological sense because

- A. It is harvested from inside the ground
B. Its pericarp is not hard and tough
C. The fruit is not as hard as the coconut
D. It is an achene

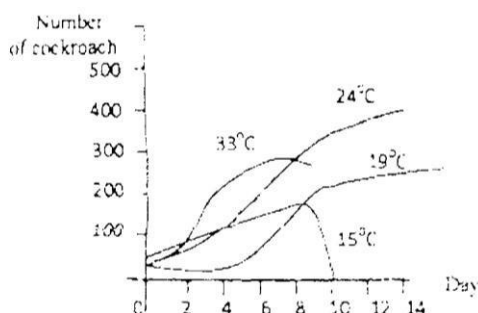
95. A photometer with a leaf shoot weighed 10.9g initially and 11.9g after 1 hour. From this it follows that:

- A. The shoot absorbed 1g of water per hour
B. 1g of water per hour evaporated from the photometer
C. The shoot lost 1g of water per hour

- D. The shoot was respiring and lost weight at a rate of 1g per hour
96. A biotic factor which affects the distribution and abundance of organisms' in a terrestrial habitat is
 A. Temperature
 B. Competition
 C. Potential hydrogen ion concentration (pH)
 D. Light
97. Mendel's second law of inheritance states that
 A. Alleles combine randomly
 B. Alleles segregate independently
 C. Alleles separate predictably
 D. Chromosomes segregate independently

36. B 37. A 38. B 39. D 40. D 41. C 42. D 43. A
 44. B 45. C 46. D 47. B 48. A 49. D
 50. A 51. A 52. D 53. B 54. C 55. C 56. D
 57. A 58. A 59. C 60. A 61. B 62. D 63. B
 64. B 65. B 66. D 67. D 68. D 69. B 70. B
 71. A 72. D 73. A 74. D 75. A 76. A 77. B 78. D
 79. A 80. D 81. A 82. A 83. D 84. D
 85. C 86. A 87. D 88. D 89. D 90. C 91.
 92. 93. B 94. B 95. A 96. B 97. B 98. C
 99. A 100. C

Use the diagram below to answer questions 98 and 99



98. The optimal temperature of breeding cockroaches is
 A. 15°C B. 19°C
 C. 24°C D. 33°C
99. At what temperature will cockroach not survive
 A. 15°C B. 19°C
 C. 24°C D. 33°C
100. The number of plant species obtained from a population study of a garden is as follows: Guinea grass (15), Ipomea spp (5), Sida spp(7) and Imperata spp (23). What is the percentage of occurrence of imperata spp
 A. 35% B. 16%
 C. 46% D. 23%

SOLUTION

1. A 2. C 3. C 4. D 5. A 6. A 7. A
 8. C 9. C 10. C 11. D 12. A 13. C 14. B
 15. A 16. A 17. C 18. C 19. A 20. C 21. A
 22. D 23. A 24. A 25. A 26. D 27. C 28. B
 29. C 30. D 31. A 32. D 33. B 34. C 35. A

POST-JAMB SCREENING TEST REVISION QUESTIONS AND ANSWERS 2015/2016 SESSION DELSU

ENGLISH LANGUAGE

Choose the word which is opposite in meaning to the underlined word in each sentence.

- Bala was put off by Ese's coquettish behaviour towards him. A. Unfriendly advances
 B. Quarrelsome attitude
 C. Inviting and friendly advances
 D. Quarrelsome attitude
- He urged other progressives in the country to join forces with his association.
 A. Radicals B. Pessimists
 C. Activists D. Conservatives
- If you are really keen on immediate results you must have to adopt this pragmatic approach
 A. Sensible B. Unrealistic
 C. Practical D. Opportunistic
- The hostess greeted her guests on a very relaxed manner.
 A. Energetic B. Athletic
 C. Stiff D. Perplexed

5. Mr. Jack was most flexible in his instructions
A. Stiff B. Upright
C. Correct D. Rigid
6. My brother's primary school education was solid and this influenced his secondary education
A. Weak B. Shaley
C. Sordid D. Sound
7. In the course of the debate, the speakers were advised to summarize their points
A. Round off B. Argue
C. Roundup D. Develop
8. He gave a painstaking account of his encounter with the armed robbers
A. Sordid B. Fearful
C. Half-hearted D. Tender-hearted.

Complete each of the following sentence by choosing the option that best suitably fills the space.

9. We cannot explain his objection ____ the new law
A. Against B. With
C. To D. For
10. The little boy is hyper active and he is therefore prone _____ accidents
A. To B. For
C. About D. With respect to
11. The boxer was highly rated _____ his doggedness and ability to absorb heavy punches
A. By B. Through
C. For D. About
12. She does not have to attend the lecture ____
A. Isn't it B. Doesn't
C. Has she D. Does she
13. We ought to stay away ____ come back
A. In case B. Should in case
C. Although D. Provided

Choose the option that conveys the meaning of the underlined portion in each of the following sentences

14. The account clerk was jailed because he cooked the books.
A. Destroyed the account books by cooking them
B. Falsified the account to his advantage
C. Sold the books in his office

- D. Set fire to the account books
15. This is miniaturized version of your house remarked the Architect
A. Exact B. Abridged
C. Decreased D. Smaller
16. The chief approached the issue with convincing disinterestedness
A. Lack of personal interest
B. Lack of interest
C. Unwillingness
D. Inability to be interested
17. The traffic situation in Lagos can lead to disastrous consequences, a man lost a very lucrative job because he was held up by
A. Delayed B. Obstructed
C. Intrigued D. Annoyed.
18. He stared at her
A. Fixed B. Peeped
C. Gazed D. Glanced
19. The general promised that the soldiers would go back to the barracks A. Military training school
B. Barricade
C. Soldiers' quarters
D. Quarter
20. Many candidates now apply to read law because it is a lucrative profession.
A. Brings quick money
B. Brings high prestige
C. Brings plenty of knowledge
D. Brings plenty of profit

INSTRUCTION: Read the passage below and answer questions 21-25.

This book consists of lectures given by me at Cambridge. Though they have been largely rewritten, I have kept a good deal of their original lecture form, as being rather less formal and less dogmatic. It is a fact that those who write on language are prone to dogmatism. I should like to make it clear at once that, if at times I put my views strongly, I do not forget that such matters of taste must remain mere matters of opinion. I have included a good many specimen passages from various authors since a book on style without abundant examples seems to me as **ineffectual** as a book on art, or biology, without abundant illustrations. Many of the passages are in French. I believe that some ability to read French

prose does seem to me most desirable "for anyone who would write well in English.

21. A suitable title for the passage is A. Original lectures given at Cambridge
B. A preface of revised version of his
C. The Cambridge lectures
D. The authentic Cambridge lectures
22. The author says that writers on language are
A. Inclined to be dogmatic
B. Opposed to dogmatism
C. Unlikely to be dogmatic
D. Ready to be dogmatic
23. To write English well, the author says it is helpful to
A. Know French thoroughly
B. Be able to write French fairly well
C. Be able to read French well D. Be able to read English well.
24. According to the author, the book is: A. A negation of his Cambridge lectures
B. An imitation of his Cambridge lectures
C. His Cambridge lectures
D. A revised version of his Cambridge lectures.
25. "Ineffectual" as used in the passage means:
A. To be efficient
B. To be incapable of producing result C. To be efficacious
D. To be lazy.

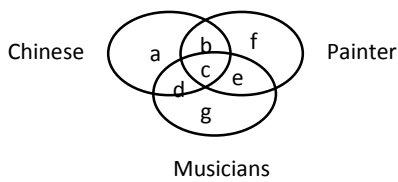
APTITUDE TEST

26. A milkman makes a profit of 20% on the sale of milk, if he were to add 10% water to the milk. By what percent would his profit increase?
A. 25% B. 60%
C. 40% D. 45%
27. A dishonest milkman tops up his bucket which is only $\frac{4}{5}$ full of milk with water. He again removes 5litres of the mixture from the bucket and adds equal quantity of water. If milk is now 60% of the mixture, what is the capacity of the bucket in litres?
A. 15 B. 20
C. 22.5 D. 25
28. A coin is tossed 5 times. What is the probability that head appears an odd number of times?
A. $\frac{2}{5}$ B. $\frac{1}{5}$
C. $\frac{1}{2}$ D. $\frac{4}{25}$

29. Ten years ago, the average age of a family of 4 members was 24 years. Two children having been born. The average age of the family is same today. What is the present youngest child if they differ in age by 2 years.
A. 1 year B. 2 years
C. 3years D. 5 years
30. The average of 8 readings is 243, out of which the average of first is 18.6 and that of the next three is 12.2. If the sixth reading is 3 less than seventh and 8 less than-eight. What is the sixth reading?
A. 24.8 B. 26.5
C. 27.6 D. 29.4
31. A Gardener want to plant 17956 trees in his garden and arranges in such way that there are as many as trees in a row. What is the total number of the trees in a row?
A. 56 B. 67
C. 47 D. 52
32. If the length and breath of rectangular plot are increased by 50% and 20% respectively, the new areas will be how many times than that of original area?
A. 4 B. $\frac{9}{6}$
C. $\frac{11}{5}$ D. $\frac{9}{5}$
33. The Average weight of a class of 29 students is 40kgs. If the weight of the teacher be included the average rises by 500grms. What is the weight of the teacher?
A. 45Kg B. 55Kg
C. 62kg D. 40 kg
34. A 10% reduction ion prices enables one to get 10kg more sugar than before for N500. What is to reduced price per kilogram of sugar?
A. N5 B. N5.45
C. N5.25 D. N5.30
35. Cost price of 20 tables is equal to the selling price of 15 tables. Then profit percent would
 $\frac{331\%}{3}$
A. $\frac{331\%}{3}$ B. 20%
C. 30% D. 45%

Direction for questions 36 to 37: In the figure given below, there are three intersecting circles each representing certain section of the people. Different regions are marked a to g. Read the statements in each of the question and choose

the letter of the region which correctly represents the statement.



36. Chinese who are painters but not musicians.
A. b B. c
C. d D. g
37. Painters who are neither Chinese nor musicians
A. b B. c
C. f D. g
38. Chinese who are musicians but not painter
A. c B. c
C. b D. a
39. Chinese who are painters as well as musicians
A. a B. b
C. c D. d
40. The tax on a commodity is diminished by 20% and its consumption increases 5%. The effect on revenue is
A. Decreases by 12% B. Decrease by 9%
C. Decrease by 8% D. None of them

Direction for questions 41 to 44: each of the following question is for alternative responses

41. Which of the following is the same as flood, fire, Cyclone?
A. Damage B. Earthquake
C. Rain D. Accident
42. Which of the following is the same as Rabbit, Rat, Mole?
A. Mongoose B. Frog
C. Earthworm D. Ant
43. Which of the following is the same as Measles, Rabies, Cancer?
A. Pneumonia B. Diabetes
C. Tetanus D. Hepatitis
44. Which of the following is the same as Wrestling, Karate, Boxing?
A. Swimming B. Judo
C. Pole Vault D. Polo

45. If clock is called television, television called oven, oven is called grinder and grinder called Iron, in what will a lady bake?
A. Radio B. Oven
C. Grinder D. Iron

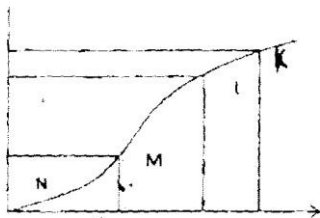
Direction:- Fill in the gap mentioned as question mark from the four options given question 46 and 50)

46. $\frac{8}{9} + \frac{3}{7} + \frac{2}{3} + ? = \frac{158}{63}$
A. $\frac{145}{21}$ B. $\frac{111}{63}$
C. $\frac{11}{21}$ D. $\frac{145}{63}$
47. $2\frac{1}{7} + 2\frac{1}{17} - ? = \frac{1}{17}$
A. $\frac{1}{17}$ B. $\frac{21}{17}$
C. $\frac{1}{17}$ D. $\frac{1}{13}$
48. On dividing a number by 999, the quotient is 366 and the remainder is 103. The number is
A. 364724 B. 365737
C. 365387 D. 366757
49. The HCF of two numbers is 11 and their LCM is 7700, If one of these numbers is 275. Then the other one is
A. 279 B. 283
C. 318 D. 308
50. The greatest possible length of scale to measure exactly 6mt 25cm, 9mt 50cm and 12mt 25cm is
A. 50cm B. 25cm
C. 125cm D. None of these
51. If the determinant of the matrix P $[4 \ 2\alpha]$ is 4, α^9 find the value of a
A. 4 B. 25
C. 25 D. $\frac{17}{3}$
52. The following are laws of indices except
A. $x^m + x^n = x^{m+n}$ B. $(x^m)^n = x^{mn}$
C. $x^0 = 1$ D. $x^m \div x^n = x^{m-n}$
53. Evaluate $\frac{4}{51} - \frac{2}{33}$

- A. $\frac{1}{2}$ B. 2
C. 3 D. $\frac{2}{3}$
54. The point of intersection of the straight lines $3x + y - 7 = 0$ and $2x - 3y - 1 = 0$ is
A. (2, 1) B. (1, 2)
C. (-2, 1) D. (-1, 3)
55. Simplify $((2\frac{2}{3} + \frac{11}{2})\frac{11}{5})/6$ A. $\frac{11}{4}$ B. $\frac{11}{6}$
C. $\frac{5}{6}$ D. $\frac{4}{5}$
56. The gradient of the straight line joining the points (5, -7) and (-2, -3) is
A. $\frac{1}{2}$ B. $\frac{2}{5}$
C. $\frac{-4}{7}$ D. $\frac{-2}{3}$
57. Bala sold an article for 6,900 and made a profit of 15%. How much did he buy it?
A. 6150 B. 5400
C. 6000 D. 7800
58. Factorize completely: $x + y - ax - ay$ A. $(x - y)(1 - a)$
B. $(x + y)(1 - a)$
C. $(x + y)(a + 1)$ D. $(x - y)(1 + a)$
59. Find the sum of the squares of 2, 5, 9
A. 65 B. 256
C. 32 D. 4
60. The interior angles of a triangle are $9x^\circ$, $4x^\circ$, and $7x^\circ$ in degrees. Therefore the angles are respectively
A. $80^\circ, 40^\circ, 60^\circ$ B. $81^\circ, 36^\circ, 63^\circ$
C. $86^\circ, 33^\circ, 61^\circ$ D. $80^\circ, 30^\circ, 70^\circ$
61. For what value of X is $\log_x 64 = 3$?
A. 16 B. 9
C. 8 D. 4
62. Find the equation whose roots are $-\frac{3}{8}$ and $\frac{1}{2}$
A. $16x^2 - 2x + 3 = 0$
B. $16x^2 - 2x + 3 = 0$
C. $16x^2 - 2x - 3 = 0$
D. $16x^2 + 2x - 3 = 0$
63. Find the sum of the first 8 terms of the GP 4, 8, 16, ...
A. 34 B. 1024
C. 512 D. 1020
64. Solve the simultaneous equations $2x - 3y = 1$ and $5x + 2y = 8$ and evaluate $x^2 - xy$
A. 16 B. 32
C. 6 D. 8
65. The angle between a plane mirror and an incident ray is 40° . What is the total angle through which the ray is deviated?
A. 40° B. 50°
C. 80° D. 100°
66. A material Q has twice the specific resistance of 'P'. A circular wire made of 'Q' has twice the diameter of a wire made of 'P'. Then for the two wires to have the same resistance, the ratio P/Q of their respective lengths must be
A. 2 B. 4
C. $\frac{1}{2}$ D. $\frac{1}{3}$
67. A 1000 watt appliance is used 10 hours a day for 8 days per month. If the cost of electricity is ₦0.10 a kwh, how much does it cost to operate the appliance for a month.
A. ₦10 B. ₦28
C. ₦67 D. ₦200
68. How much heat is given out when a lump of aluminum of mass 200g and specific heat capacity $880 \text{ J/kg}^\circ\text{C}$ cool from 60°C to 20°C
A. 5411J B. 6513J
C. 7040J D. 8130J
69. In which of the following medium do sound wave have the greatest speed A. Vacuum B. Air
C. Water D. Steel
70. The distance between two consecutive antinodes is
A. $\sqrt{1}$ B. $\sqrt{2}$
C. $\sqrt{3}$ D. $\sqrt{4}$
71. A captain directs a ship due north at 100 Km/h while the river is flowing due east at 2.5 Km/h.

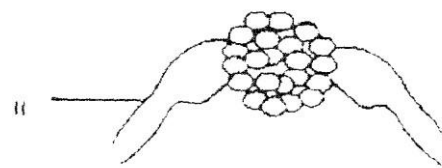
- What is the resultant velocity of the ship?
- A. 103 km/h B. 144km/h
C. 171 km/h D. 194km/h
72. Which of the following is the reason while water is not suitable for use in thermometers?
- A. It is not opaque
B. It is a poor conductor
C. It wet's glass
D. All of these
73. A metal whose linear expansivity is $0.000011 \text{ } ^\circ\text{C}^{-1}$ has a length of 54.22 cm at 21°C . At what temperature will its length become 54.38cm?
- A. 268.27°C B. 271.27°C
C. 279.27°C D. 289.27°C
74. The acceleration of a satellite of mass m in a circular orbit of radius R around a planet of mass M is equal to which of the following??
- A. $G\frac{M}{R^2}$ B. $G\frac{mM}{R^2}$
C. $G\frac{m}{R^2}$ D. $G\frac{mM}{R}$
75. An object of mass 6kg moves round a circle of radius 3.5m with a speed of 9m/s. What is its angular velocity?
- A. 1.55 rad/s B. 2.57 rad/s C. 5.5 rad/s D. 9.0 rad/s
76. A crane lifts a load of weight 54500 N through a height of 15 in 10 seconds. What is the power of the crane.
- A. 3350W B. 4225W
C. 6750W D. 81750W
77. The gas that can be dried using concentrated tetraoxosulphate (vi) acid is
- A. Hydrogen bromide B. Sulphur (iv) oxide
C. Hydrogen sulphide D. Ammonia
78. Which of the following equimolar solutions will have the highest pH
- A. CH_3COONa B. NaOH
C. HCl D. NH_4OH .
79. The repeating unit in natural rubber is
- A. Isoprene B. N-propene
C. Alkyne D. Neoprene
80. When 10g of sodium hydroxide is dissolved in 1000cm³ of water, the solution approximately
- A. 0.01mol dm^{-3}
B. 0.10mol dm^{-3}
C. 0.25mol dm^{-3}
D. 0.50mol dm^{-3} (Na=23, H=1, O=16)
81. A compound contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen. If the molar mass of the compound is 180, find molecular formula
- A. CH_2O B. $\text{C}_3\text{H}_6\text{O}_3$,
C. $\text{C}_6\text{H}_{12}\text{O}$ D. $\text{C}_6\text{H}_6\text{O}_3$.
82. Sulphur (iv) oxide bleaches by
- A. Reduction B. Oxidation
C. Hydration D. Absorption
83. What amount of mercury would be liberated the same quantity of electricity that liberates 0.65g of zinc is applied?
- A. 2.01g B. 1.00g
C. 4.02g D. 8.04g
84. Which of the following gives a precipitate when treated with NaOH solution?
- A. AlCl_3 B. NH_4Cl
C. CH_3COONa D. Na_2CO_3
85. What volume of oxygen is produced from the decomposition of 2 moles of KNO_3 at s.t.p?
- A. 22.4dm^3 B. 33.6dm^3
C. 44.8dm^3 D. 67.2dm^3
86. In the industrial production of hydrogen from methane, the carbon (iv) oxide produced along with hydrogen gas is removed by
- A. Washing under pressure
B. Drying over phosphorus (v) oxide
C. Passing the mixture into lime water using ammoniacal copper (i) chloride
D. None of the above
87. How long would it take to deposit 0.08g of copper from CuCl_2 solution by passing a current of 0.5A?
- A. 6mins B. 8mins
C. 24mins D. 48mins
88. The conversion of open chain alkenes into cycloalkanes and aromatic compounds is called
- A. Cracking B. Reforming
C. Isomerisation D. Polymerization

89. In epigeal germination, the cotyledons are:
- Carried above the ground by the elongating hypocotyls
 - Pulled underground by the elongating hypocotyls
 - Pulled underground by the elongating epicotyls
 - Carried above the ground by the elongating epicotyls



90. Which part of the curve represent the fastest rate of growth?
- K
 - L
 - M
 - N
91. The grand period of growth as represented by
- N
 - M
 - L
 - D. K
92. The anatomical evidence usually used in support of evolutionary relationship among whales, humans, birds and dogs is the possession of
- Thick skin
 - Pentadactyl limb
 - Tail
 - Epidennal structure
93. The factors that determines the distribution of vegetation zones are:
- Temperature, light, air and humidity
 - Light, humidity, air and mist
 - Temperature, light, rain and humidity
 - Humidity, snow, frost and dew
94. Heredity variation is best described as
- Characters transmitted from generation to generation
 - Traits acquired from diseases
 - Characters that show up due to social
 - Traits acquired from habits and environmental conditions.
95. The autonomic nervous system comprise
- Sympathetic and parasympathetic systems
 - Cranial and sacral nerves
 - Brain and cranial nerves
 - Spinal cord and spinal nerves

Using the diagram below, answer questions 96 and 97



96. The structure labelled (i) is formed as a result of the fusion of
- Two pairs of nuclei
 - Several pairs of nuclei
 - A pair of nuclei
 - Two pairs of nuclei
97. The special name of the part labelled II is
- Gametagium
 - Hypha
 - Suspensor
 - Zygospore
98. During blood transfusion, agglutination may occur as a result of the reaction between
- Similar antigens and antibodies
 - Contrasting antigens and antibodies
 - Two different antigens
 - Two different antibodies
99. Which of the following is not part of the objectives of conservation of natural resources?
- To eliminate wild animals in order to enable other plants and animals survive
 - To use our resources on sustained yield
 - To guard against indiscriminate killing of plants and animals
 - To guard against the extinction and exhaustion of natural resources
100. Assuming A is gene for normal skin and it is dominant while a is gene for albinism and it is recessive. What is the likely genotype of a couple who had 50% normal 50% albino
- AA, Aa
 - AA, aa
 - Aa, Aa
 - Aa, aa

SOLUTION

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. B | 4. C | 5. D | 6. A | 7. D |
| 8. C | 9. C | 10. A | 11. C | 12. D | 13. A | 14. B |
| 15. D | 16. A | 17. A | 18. C | 19. C | 20. D | 21. B |
| 22. A | 23. C | 24. D | 25. B | 26. C | 27. C | 28. D |
| 29. C | 30. D | 31. B | 32. D | 33. B | 34. A | 35. A |
| 36. A | 37. C | 38. B | 39. C | 40. A | 41. B | 42. A |
| 43. D | 44. B | 45. C | 46. B | 47. B | 48. B | 49. B |
| 50. D | 51. A | 52. D | 53. B | 54. A | 55. C | 56. C |
| 57. C | 58. A | 59. B | 60. D | 61. D | 62. C | 63. D |
| 64. C | 65. C | 66. C | 67. B | 68. C | 69. D | 70. B |
| 71. A | 72. D | 73. D | 74. A | 75. B | 76. D | 77. B |

78. B 79. A 80. C 81. C 82. B 83. A 84. A
 85. B 86. C 87. B 88. B 89. A 90. C 91. A
 92. B 93. C 94. A 95. A 96. A 97. C 98. B
 99. A 100. D

POST-JAMB SCREENING TEST

REVISION QUESTIONS AND ANSWERS

2014/2015 SESSION DELSU

Fill in the missing numbers in questions 1-2

1. 100,98,94,88,80,70,? A. 58 B. 48 C. 68
 D. 64

2. 1.5, 3.5, 5.9, 13.5, ?
 A. 17 B. 19 C. 5 D. 21

3. Bill and Ben share flower pots in the ratio of 3:5. If Bill has 180 flower pots, how many has Ben?
 A. 200 B. 350
 C. 300 D. 150

4. How many birthdays does the average man have?
 A. 0 B. 1 C. 70 D. 65

5. What is a percent of b divided by b percent of a?
 A. a B. b C. 1 D. 100

6. Five farmers have 7,9,11,13, & 14 apple trees respectively in their orchards. Last Year, each of them discovered that every tree in their own orchard bear exactly the same number of apples. Further, if the third farmer gives one apple to the first and the fifth gives three to each of the second and the fourth, they would all have exactly the same number of apples. What were the yields per tree in the orchards of the third and fourth farmers?
 A. 4 & 6 B. 10 & 7
 C. 11 & 9 D. 16 & 10

7. Five boys were climbing a hill. J was following H. R was just ahead of G K was between G & H. They were climbing up in a column. Who was the second?
 A. H B. G C. K D. R

8. Which of the following fractions is less than $\frac{1}{3}$?
 A. $\frac{22}{62}$ B. $\frac{15}{46}$
 C. $\frac{2}{3}$ D. 1

9. What is the smallest number by which 2880 must be divided in order to put it into a perfect square?

A. 3 B. 4 C. 5 D. 6

The question 10-11 are based on the following pattern. The problems below contain a question and two statements giving certain data. You have to decide whether the data given in the statements are sufficient for answering the questions.

- (a) If statement (I) alone is sufficient but statement (II) alone is not sufficient.
- (b) If statement (II) alone is sufficient but statement (I) alone is not sufficient.
- (c) If both statements together are sufficient but neither of statements alone is sufficient.
- (d) If both together are not sufficient.

10. Is $A = c$? A, B, C are real numbers
 (i) $A - B = B - C$ (ii) $A - 2C = C - 2B$

11. Was A. Vinash early, on time or late for work?
 (I) He thought his watch was 10 minutes fast
 (II) Actually his watch was 5 minutes slow

12. Of Abdul, Binoy, and Chandini:
 (i) Each member belongs to Tee family whose members always tell the truth or to the El family whose members always
 (ii) Abdul says "Either I belong or Binoy belongs to a different family from the other two."
 A. Abdul's family-Tee
 B. Chandini's family-El
 C. Binoy's family-El
 D. Tee's family

13. OBLIGATORY (Give synonym)
 A. Useful B. Required
 C. Stubborn D. Agreeable

14. KEEN (Give antonym)
 A. Dull B. Rogue
 C. Ardent D. Shrewd

15. How many two cent stamps are there in a dozen?
 A. 0 B. 1 C. 6 D. 12

16. Mal is one-and-a-half times as old as Sal, and Sal is one-and-a-half times as old as Al. How old are Al, Sal and Mal if their combined ages total 114
 A. 24, 30 and 54 B. 24, 36 and 54
 C. 36, 54 and 65 D. 12, 24 and 36

17. A doctor gives you three pills telling you to take one every half hour. How many minutes would the pills last?
A. 0 B. 30 C. 60 D.90

18. Mary's father has five daughters: 1. Nana, 2. Nene, 3. Nini, 4. Nono. What is the name of the fifth daughter?
A. Don'tknow B. Nano
C. Mary D. Nunu

Direction 19-20

John is undecided which of the four novels to buy. He is considering a spy thriller, a Murder mystery, a Gothic romance and a science fiction novel. The books are written by Rothko, Gorky, Burchfield and Hopper, not necessary in that order, and published by Heron, Piegon, Blueja and sparrow not necessary in that order.

- (i) The book by Rothko is published by Sparrow
(ii) The spy thriller is published by Heron
(iii) The science fiction novel is by Burchfield and is not published by Blueja
(iv) The Gothic romance is by Hopper.

19. Piegon publishes_____ A. Science fiction
B. Spy thriller
C. Murder mystery D. Gothic romance

20. The novel by Gordy is_____ A. Science fiction
B. Spy thriller
C. Murder mystery D. Gothic romance

21. A man walks east and turns right and then from there to his left and then 45 degrees to his right. In which direction did he go?
A. North cast B. North south
C. Southeast D. Northwest

22. A man named Nepo say's I am three times my son's age. My father is 40 years more than twice my age. Together the three of us are a mere 1240 years old. How old is Nepo?
A. 336 B. 360
C.330 D.660

23. One dog tells the other "there are two dogs in front of me". The other one also shouts that he too had two behind him. How many are they?
A.6 B. 5 C. 4 D.3 24. If in a certain code 'RANGE' is coded as 12345 and 'RANDOM' is code as 123678. Then the code for the word 'MANGO' would be

- A. 82357 B. 89343
C. 84629 D. 82347

25. One of Mr. Horton, his wife, their son, and Mr. Horton's mother is a doctor a doctor and another is a lawyer (i) If the doctor is younger than the lawyer, then the doctor and the lawyer are not blood relatives (ii) If the doctor is a woman, then the doctor and the lawyer are blood relatives (iii) If the lawyer is a man, then the doctor is a man. Whose occupation you know?
A. Mr. Horton is the Lawyer
B. Mr. Horton is a Doctor
C. Mr. Horton's Son is the Doctor
D. Mr. Horton's Son is the Doctor

ENGLISH LANGUAGE

In each question 26-30 choose the most appropriate option opposite in meaning to the word(s) underlined.

26. This is an abridge edition of Oliver Twist
A. Reprinted B. Enlarged
C. Extinct D. Banned
27. Good students can easily identify spurious arguments
A. Genuine B. False
C. illogical D. Interest
28. Our party is wielding potent political power
A. Assuming B. Renouncing
C. restoring D. Reappraising
30. The severity of the harmattan helped me complete my assignment in record time because I had no choice but to lock myself indoors
A. Mildness B. Moderation
C. Gentility D. Enmity

In each of the question 31-35 select the option that best explains the information conveyed in the sentence

31. The politician's inflammatory remarks were heavily criticized
A. The politician's tribalistic remarks were heavily criticized
B. The politician's provocative remarks were heavily criticized
C. The politician's bombastic remarks were heavily criticized

- D. The politician's rude remarks were heavily criticized.
32. If he were here it could have been more fun
 A. There was no fun because he was not present
 B. He did not show up and so the occasion lack much fun
 C. He was been expected to supply more fun
 D. He was expected but did not show up to liven the audience
33. Though Shola is our elected representative, he often takes a rather jaundice view of our problem
 A. He takes a rather lazy view of our problem
 B. He takes a rather sickly view of our problem
 C. He takes a rather forceful view of our problem
 D. He takes an unfavorable position concerning our problem
34. Toritseju is traveling tomorrow A. Toritseju may travel tomorrow
 B. Toritseju has to travel tomorrow
 C. Toritseju will travel tomorrow
 D. Toritseju should travel tomorrow
35. Kate would make a mess of cooking the yam
 A. Kate cannot cook
 B. Kate cooks yam well
 C. Kate will not cook the yam well
 D. Kate does not like cooking yam

Instruction: Read the passage below and answer the question that follow

Professor 1km emerged from the chain incident a changed man. During the preceding months when his wife has talked persistently about Dr. Okoro and his American friends he had paid little attention to her. He assumed that nagging was her way of life. Dr. Okoro did not constitute a threat to him; they were not in the same department so they could not be competing for the headship of the department. Even if they taught the same subject, Okoro could surely not be deemed to be a rival to him. He was an associate professor while Okoro had only just become a lecturer grade II. If he had yielded to his father's pressure to marry early, he might have produced a child as old as Dr. Okoro. Two members of the provisional council has intentionally dropped broad hints that he was lined up to take over from Prof. Wilson as Vice-Chancellor. He did not therefore need to take any notice of Dr. Okoro's attitude towards him than a cow takes notice of a fly perching on its back.

36. From this passage, it is clear that professor 1km

- A. loves his wife very dearly
 B. does not value what his wife says
 C. hates his wife
 D. does not trust his wife
37. Appointment of a Vice-Chancellor is done by
 A. lecturers in the university
 B. Professors in the university
 C. members of the Provisional Council
 D. The president
38. Persistently as used in the passage means
 A. Not discouraged in a course of action
 B. Discouraged by previous failure in a course of action
 C. Vacillating in a course of action
 D. Infirm in a course of action
39. From the passage it is clear that Professor Ikin
 A. Married at an early age
 B. Had children at an early age
 C. Married early but did not produce children immediately after marriage.
 D. Did not marry at an early age.
40. Dr. Okoro cannot be seen to be a threat to Professor Ikin because
 A. A cow does not take notice of a fly perching on its back.
 B. He is very junior to Professor Ikin in the university hierarchy
 C. Professor Ikin is old enough to be Dr Okoro's father
 D. A Doctor is inferior to a professor

In each of the question 41-45, choose the option nearest in meaning to the word or phrase underlined

41. The Vice-Chancellor appointed a commission of inquiry to go into the students complaint carefully and without prejudice
 A. ascertain B. investigate
 C. search D. account for
42. Many Nigerians have inured themselves to hardship
 A. subjected B. Submitted
 C. accustomed D. Endeared
43. Akpos plans for the future are in the air
 A. suspended B. Uncertain

C. empty

D. Floating

A. 48 B. 60 C. 72 D.36

44. In the match against Rangers, Warn seaisiders turned to be the dark horse A. played most brilliantly
B. played below their usual form
C. won unexpectedly
D. won as expected

45. The bill has to wait as we are insolvent now
A. insoluble B. Affluent
C. unsettled D. Bankrupt

The passage has gaps numbered 46 -50. Immediately following each gap four options are provided, choose the most appropriate option for each gap.

In today's society, few people at home feel __46__ [(A) estranged (B) discouraged (C) had (D) intrigued] from other people and __47__ (A) cut out (B) cut off (C) cut away (D) cut down] from the world about them. Telephone, television car, common social and educational __48__ [(A) interest (B) associations (C) interests (D) escapades] with other parents, all keep married couples in living __49__ [(A) contract (B) contrast (C) activity (D) together] with the rest of the community. If a person is __50__ [(A) deprived (B) isolated (C) neglected (D) cheated].

GENERAL PAPER

51. Find the length of arc of circle of radius 6cm if it subtends angle 108° at the Centre. Take $\pi = 3.14$
A. 24.8cm B. 34.8cm
C. 11.3cm D. 8.91cm
52. The score of 5 pupils in a test are 3,4,2,x,3. If the average score is 2.9, find x.
A. 1.5 B. 2.5
C. 3.5 D. 4.5
53. There are 5 blue, 7 green and 3 red balls in a bag. A ball is drawn from the bag at random. What is the probability that it is not green?
A. $\frac{8}{15}$ B. $\frac{7}{15}$
C. $\frac{3}{110}$ D. $\frac{1}{3}$

54. The gradient of the curve $y = 5x^3 + 3x^2$ at $x=2$ is

55. If the sum of n terms of the A.D. 3,10,17,... is 2730, find n
A. 28 B. 37 C. 18 D. 42

56. The length of a rectangle is 2m greater than the breadth. If the area of the rectangle is 168m^2 , find its length.
A. 19m B. 11m
C. 14m D. 16m

57. Factorize $1-16x^4$
A. $(1+4x)(1-4x)$ B. $(4x-4x)$
C. $(1+4x^2)(1-2x)$ D. $4x(1-4x^3)$

58. Solve for x if $32x = 0.25$
A. $-\frac{2}{5}$ B. -3
C. $-\frac{4}{7}$ D. $-\frac{1}{2}$

59. Given that the number 48×81 is divided by 9; then the value of ϕ^* is
A. 5 B. 6 C. 8 D. 9

60. Let the universal set $U = \{x: -3 < x \leq 4\}$ and its subsets $A = \{-1, 1, 2\}$ $B = \{x: 1 \leq x \leq 3\}$ Find $A \cup B$
A. $\{0, 2, 3\}$ B. $\{-1, 2\}$
C. $\{-4, -3, 2\}$ D. $\{-1, 1, 2, 3\}$

61. The remainder on dividing $x^3 - 6x + 1$ by $x-3$ is
A. 12 B. 21 C. 10 D. 16

62. Which of the following quadratic equations has as $x = -2$ or $x = 7$
A. $x^2 + 2x - 7 = 0$ B. $x^2 - 2x + 7 = 0$
C. $x^2 - 5x + 14 = 0$ D. $x^2 - 5x - 14 = 0$

63. Find the area of a trapezium whose parallel sides are 8cm and 6cm given that its height is 9cm.
A. 24cm^2 B. 63cm^2
C. 58cm^2 D. 46cm^2

64. The percentage of water of crystallization in $\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$ is
A. 33% B. 44%
C. 55% D. 87%
(Zn=65, S=32, O=16, H=1)

65. Metals can be extracted from their ores by a process involving
A. Oxidation B. Reduction
C. Decomposition D. Hydrolysis.
66. The pressure of 100cm^3 of oxygen at 35°C is 750mmHg . What will be the volume of the gas if the pressure is reduced to 100mmHg without changing temperature
A. 650cm^3 B. 850cm^3
C. 580cm^3 D. 750cm^3
67. How many structural isomers of chlorobutane are possible?
A. 1 B. 2 C. 3 D. 4
68. A compound gives an orange red colour to a non luminous flame. This compound is likely to contain
A. Na^+ B. Ca^{2+}
C. Fe^{2+} D. Fe^{3+}
69. The IUPAC nomenclature of the compound LiAlH_4 is
A. Lithium aluminium hydride
B. Tetrahydrido lithium aluminate(III)
C. Aluminium tetrahydrido lithium
D. Lithium tetrahydridaluminatate(III)
70. What is the molecular formula of a compound with empirical formula CH_2O and vapour density of 90? (H=1, C= 12, O 16)
A. $\text{C}_4\text{H}_8\text{O}_2$ B. $\text{C}_4\text{H}_8\text{O}_3$
C. $\text{C}_6\text{H}_{10}\text{O}_5$ D. $\text{C}_6\text{H}_{12}\text{O}_6$
71. Which of the following metals exists as liquid at ordinary temperature.
A. Copper B. Gold
C. Mercury D. Silver
72. The basicity of CH_3COOH is
A. 1 B. 2 C. 3 D. 4
73. The entropy and enthalpy of a system are a measure of
A. degree of disorderliness and heat content respectively
B. heat content and degree of disorderliness respectively
C. heat content of a system only
D. degree of disorderliness only
74. Destructive distillation of coal means A. heating coal in plentiful supply of air
B. burning coal in air to produce water
C. heating coal in the absence of air
D. Separating coal into various allotropes of carbon
75. The colour of methyl orange in alkaline is A. Pink
B. Orange
C. Red D. Yellow.
76. Permanent hardness of water can be removed by
A. Boiling
B. Adding of slake lime
C. Filtration
D. Adding caustic soda.
77. In a circular motion, the force which is directed at the centre is referred to as
A. Centrifugal force B. Central force
C. Circular force D. Centripetal force
78. Which of the following instruments is used to measure the degree of hotness or coldness of a body?
A. Temperature B. Thermometer
C. Glass tube D. Oven
79. The reciprocal of wavelength is called... A. Wave frequency B. Wave velocity
C. Wave amplitude D. Wave number
80. Which of the following is an application of rectilinear propagation of light?
A. Eclipse B. Shadows
C. Pinhole camera D. Inertia
81. _____ is for rotational motion as mass is for linear motion
A. mass B. Torque
C. moment of inertia D. Inertia
82. Consider an object undergoing motion under a force. If the mass and the force are doubled, then the acceleration will
A. Double B. Become thrice
C. Become four times D. Remain the same
83. The temperature difference between the ice point and the steam point in a Thermometer is called
A. Fundamental interval
B. Temperature scale
C. Degree

- D. Fixed point

84. A 1.5kg piece of aluminium at 210°C is dropped into 5kg of water at 10°C . What will be the final temperature of the aluminium if the specific heat capacity of aluminium and water are respectively $880\text{J/kg}^{\circ}\text{C}$ and $4200\text{J/kg}^{\circ}\text{C}$

A. 22° B. 33°C
C. 42°C D. 56°C

85. 1KWh is equal to A. 746W B. $3.6 \times 10^6\text{J}$
C. 3.6hp D. 746kW

86. What is the effective capacitance of three capacitors of capacitance 3F, 4F and 5F if they are connected in series?

A. 1.28F B. 2.00F
C. 3.00F D. 12.00F

87. Which of the following pairs are measured in the same units?

A. Force and Energy
B. Momentum and Work
C. Power and Energy
D. Impulse and Momentum

88. A car moving at a velocity of 60m/s, accelerates uniformly at a rate of 5m/s^2 until it attained a velocity of 120m/s. What is the distance covered?

A. 1080m B. 720m
C. 660m D. 420m

89. Which of the following about sex determination is correct? At fertilization.

A. The probability of male to female offspring is $\frac{1}{2}$
B. The Y chromosomes of the father decreases the probability of having a male child to $\frac{3}{4}$
C. The XX chromosomes of the mother decreases the probability of having a female child to $\frac{3}{4}$
D. The Vigour of either parent determines the sex of the offspring

90. The hepatic portal vein is characterized by

A. beginning and ending with capillaries
B. connecting the liver and kidney
C. Carrying deoxygenated blood
D. Ending with large capillaries

91. $2\text{H}_2\text{O} \rightarrow 2\text{H}_2 + \text{O}_2$. The equation represents a part of the light stage of photosynthesis. Which of the following must be present for their reaction to occur?

A. Enzyme and light energy
B. Carbon dioxide and light energy
C. Light energy and chlorophyll
D. Chlorophyll and enzyme

92. Given that a man who has the trait for colour blindness marries a normal woman. What percentage of their children would be sufferers, carriers and normal respectively.

A. 25%, 25% and 50%
B. 25%, 50% and 25%
C. 25%, 37.5% and 37.5%
D. 50%, 25% and 25%

93. Oil is usually sprayed over stagnant water in which mosquito breed in order to

A. Provide landing surface for the adult
B. Prevent larva or pupa from attaching itself to the surface
C. Prevent the water from overheating
D. Lubricate the larva for easy movement

94. In a cross involving a heterozygous red flowered plant (Rr) and white flowered plant (rr), what is the probability that the offspring will be Br?

A. 1 B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{3}{4}$

95. The maize grain is regarded as a fruit not a seed because

A. It is covered by a sheet of leaves
B. The testa and the fruit wall fuse after fertilization.
C. It has both endosperm and cotyledon
D. The pericarp and seed coat are separate

96. Oxygen from the atmosphere reaches the mammalian blood stream through one of the paths indicated below. Indicate the correct path.
A = mouth B = trachea C = bronchiole D = Nostril
E = Bronchus F = Alveol

A. $a \rightarrow b \rightarrow c \rightarrow f \rightarrow d$ B. $b \rightarrow \rightarrow \rightarrow \rightarrow d \rightarrow f$
C. $d \rightarrow \rightarrow c \rightarrow \rightarrow b$ D. $f \rightarrow \rightarrow \rightarrow \rightarrow$
E. $e \rightarrow c \rightarrow d$ F. $c \rightarrow f$

A. i and ii only B. i and iii only
C. i and iv only D. i, iii and iv only

97. Conservation of wild life is necessary mainly because
- Wild species cannot resist disease better than domesticated species
 - Predator on wild animals help in the control of pests
 - Many people derive pleasure from observing wild animals
 - It helps to prevent extinction of species
98. The role of Golgi complex in a eukaryotic cell is to
- Conduct ions in and out of the cell
 - Transport genetic materials out of the cell
 - Provide attachment for ribosomal granules
 - Transport organic materials in and out of the cell

Use the information below on factors required during photosynthesis to answer question 99 and

100 (i) CO_2 (ii) Chlorophyll (iii) Water (iv) ATP (v) NADPH_2

99. Which of the above factors can be a limiting factor for the process of photosynthesis to proceed?
100. The factor(s) required for the dark phase of photosynthesis are
- iii and iv only
 - iii and v only
 - iii, iv and v only
 - i, iii and iv only.

SOLUTION

- | | | | | | | |
|-------|--------|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. C | 4. B | 5. C | 6. C | 7. B |
| 8. B | 9. C | 10. C | 11. D | 12. C | 13. D | 14. A |
| 15. D | 16. B | 17. C | 18. C | 19. A | 20. B | 21. D |
| 22. B | 23. D | 24. D | 25. B | 26. B | 27. A | 28. B |
| 29. A | 30. A | 31. B | 32. D | 33. D | 34. C | 35. C |
| 36. B | 37. C | 38. A | 39. D | 40. B | 41. B | 42. C |
| 43. B | 44. C | 45. D | 46. A | 47. B | 48. C | 49. A |
| 50. B | 51. C | 52. B | 53. A | 54. C | 55. A | 56. C |
| 57. C | 58. A | 59. B | 60. D | 61. C | 62. D | 63. B |
| 64. B | 65. B | 66. D | 67. B | 68. B | 69. A | 70. D |
| 71. C | 72. A | 73. A | 74. C | 75. D | 76. D | 77. D |
| 78. B | 79. D | 80. D | 81. C | 82. D | 83. A | 84. A |
| 85. B | 86. A | 87. D | 88. A | 89. A | 90. A | 91. C |
| 92. C | 93. B | 94. A | 95. B | 96. C | 97. D | 98. D |
| 99. A | 100. C | | | | | |

**POST-JAMB SCREENING TEST
REVISION QUESTIONS AND ANSWERS
2014/2015 SESSION DELSU**

APTITUDE

1. The number that does not have a reciprocal is ____
A. infinity B. 1
C. 0 D. None of the above
2. How big will an angle of one and a half degree look through a glass that magnifies things three times?
A. $1\frac{1}{2}$ degree B. $\frac{9}{2}$ degrees
C. $\frac{7}{2}$ degree D. $3\frac{1}{2}$ degrees
3. A shopkeeper bought a watch for N400 and sold it for N500. What is his profit percentage?
A. 100% B. 20%
C. 25% D. 10%
4. A number 54 is divided into two parts, so that 10 times the first and 22 times second are together equal to 780. The difference of the
A. 12 B. 14 C. 20 D. 34
5. DISPARITY (Give Synonym) A. Injustice
B. Unlikeness
C. Partiality D. Distortion
6. SALVAGE (Give antonym)
A. Outfit B. Lose
C. Bum D. Remove
7. 2,5,75,_____, 13.25,17,20.75
A. 8.5 B. 9.5 C. 6.75 D. 10.5
8. A man was engaged on a job for 30 days on the condition that he would get a wage of ₦10 for the day he works, but he has to pay a fine of ₦2 for each day of his absence. If he gets ₦216 at the end, he was absent for work for _____ days
A. 3 days B. 8 days
C. 5 days D. 7 days

For questions 9 and 10, find the next term in the sequence 9. 6, 34, 60, 120, 210

- A. 336 B. 366
C. 330 D. 660
10. 8,24,12,36,18,54,_____ A. 33 B. 66 C. 27 D. 30

For question 11 - 12, fill in the missing number in each sequence

11. 0, 1, 4, 9, 16, 25,36,49,? A. 64 B. 58 C. 76 D. 88
12. 9, 18, 27,?,45,?36 A. 28,60 B. 36, 54
C. 37,58 D. 36,60
13. I drove 60km at 30kmph and then an additional 60km at 50 kmph. Compute my average speed over my 120km.
A. 35.5 B. 39.5
C. 37.5 D. 29
14. Father's age is 5 times his son's age. 4 years back the father was 9 times older than son. Find the father's present age.
A. 45 years B. 40Years
C. 50years D. 35 years
15. In simple interest what sum amounts of ₦1120 In 4years and ₦1200 in 5years?
A. ₦500 B. ₦600
C. ₦800 D. ₦900
16. Which of the following statements drawn from the given statements are correct? Given: All watches sold in that shop are of high standard. Some of the HMT watches are sold in that shop
A. All watches of high standard were manufactured by HMT
B. Some of the HMT watches are of high standard
C. None of the HMT watches are of high standard
D. Some of the HMT watches of high standard are sold in that shop
17. 0,100, 6,94,12,88,18,82,_____,_____
A. 24,76 B. 24,66
C. 28,56 D. 36,78
18. 1,1,2,_____, 24, 120,720,
A. 16 B. 8 C.10 D.6
19. If point P is on line segment AB, then which of the following is always true?
A. AP=PB B. AP>PB
C. PB>AP D. AB>AP
20. All men are vertebrates. Some mammals are vertebrates. Which of the following conclusions draw from the

above statement is correct? A. All men are mammals

B. All mammals are men

C. Some vertebrates are mammal

D. None

21. Net: Ball::

A. Winter: Weather

B. Game: Penant

C. Hook: Fish

D. Stadium: Seats

22. Kick: Football::?

A. Manager: Team

B. Break: Pieces

C. Smoke: Pipe

D. Kill: Bomb

23. Integrity of character, Honesty, dependability and discipline___ with a genuine interest in your work will got a long way in the achievement of success in your professional life.

A. Coupled

B. Adjoined

C. Fixed

D. Attached

Direction for questions 24-25

John is undecided which of the four novels to buy. He is considering a spy thriller a Murder mystery, a Gothic romance and a science fiction novel. The books are written by Rothko, Gorky, Burchfield and Hopper not necessary in that order

24. John purchases books by the authors whose names come first and third in alphabetical order. He does not buy the books___

25. On the basis of the first paragraph (2), (3) and (4) only, it is possible to reduce that

i. Rothko wrote the murder mystery or the spy thriller

ii. Sparrow published the or the spy mystery or the spy thriller

iii. The book by Burchfield is published by Sparrow which of the following statement is not true

A. iii B. ii C. i D. all of them

ENGLISH LANGUAGE

In each questions 26 - 30 choose the most appropriate option opposite in meaning to the word (s) underlined.

26. The increase in transport fares deterred our club from planning an excursion this year

A. Deferred

B. Irritated

C. Encouraged

D. Restricted

27. This card entitles you to attend the film show

A. Disqualifies

B. Discourages

C. Satisfies

D. Proclaims

28. We will surely y the idea of introducing a new product.

A. Accept

B. Reject

C. Refuse

D. Think about

29. Ochuko is naturally taciturn

A. Friendly

B. Cheerful

C. Garrulous

D. Lively

30. The executive governor upheld the decision of the cabinet

A. held up

B. Abolished

C. undercut

D. Reversed

Complete each of the following sentences by choosing the option that most suitably fills the space

31. I am sorry ___ to attend the meeting yesterday

A. For failure

B. In Falling

C. To having failed

D. for failing

32. It all depended on what___ A. Did he want

B. He wanted

C. Dose he Want

D. did he want

33. My colleagues ___before I arrived A. Have been writing

B. Started writing

C. Wrote

D. Had started writing

34. John reported to his manager that he _____ oversees

A. Has never been

B. Had never been

C. Never

D. Is never

35. I started writing this test at 8:00am. It is 10:00am now by 11:00am, I _____ it for three

A. Shall be writing

B. Have been writing

C. Shall have been writing

D. Am writing

Instruction: Read the passage below and answer the questions that follow: My good people, I come before you this week as a man whose honesty and integrity have been questioned. Now, the usual political thing to do when charges are levelled against you is either to ignore them without giving details is or provide information. However before I answer any

of your questions, let me state categorically that I have not touched a kobo of the ₦50,000 we contributed. Every kobo of it has been used in defraying political campaign expenses.

As a matter of fact, during one of my “meet the press” conferences, Tony Agiwa accosted me and said, ‘Honourable senator what about this fund we hear about? I told him there was no secret about the fund and that he should meet Abu O. Abu to get details of the fund. I told him, “you will find that the purpose of the fund was primarily to defray political expenses” In answer to another one of questions, I said that neither contributors to the fund, nor contributors to any of my campaigns has ever received any special consideration that he would not have received as an ordinary constituent. And I can say that never, since I became a senator, have I made a telephone call for them to an agency or have I gone down to an agency on their behalf. Records will show that and these records are in the hands of the Administrators

36. Accosted in the passage means A. Grossly insulted B. Greeted C. Fought with D. Forced
37. The author is A. Exonerating himself from allegations of embezzlement B. Ignoring the allegations of embezzlement C. Taking part in a political campaign D. Trying to implicate Abu O. Abu, a fellow politician
38. Constituent in the passage means A. Section of the constituency B. His entire constituency C. A supporter D. A person having voting right where he
39. Abu O. Abu is portrayed as A. A corrupt politician B. The administrator of the fund C. An ordinary tax payer D. A government official
40. It appears that the 50,000 A. Has been spent B. Has been paid into the government treasury C. Will not be paid to government treasury D. Will be used for future campaign

In each of the questions 41 - 45, choose the option nearest in meaning to the word or phrase underlined’

41. All John’s plans fell through
A. Failed B. were accomplished
C. were rejected D. Fell
42. The cost of the experiment represents a break through in medical science A. A colossal failure
B. A breaking point
C. Catastrophe
D. an outstanding success
43. I have been able to observe Ese at close quarters
A. In government houses nearby
B. Within a short space of time
C. At close range
D. At regular intervals
44. Ovie spoke with this heart in his mouth
A. Courageously
B. With such unusual cowardice
C. With a lot of confusion in his speech
D. With fright and agitation
45. From the way Shade talks, you can see that she is such a bore
A. Uninteresting B. Brilliant
C. Humorous D. Rude

In each of the questions 46-50 select the option that best explains the information conveyed in the sentence.

46. The manager said the new loaf is the last word in bakery
A. the loaf was the last in the baker’s directory
B. the loaf was the best ever baked
C. the loaf was the last to be baked D. the loaf was the worst to be baked.
47. The horizon may not be clear now, but the troubled spots have certainly been noted
A. The horizon dependent on the troubled spots
B. Clarity of the horizon is dependent on the perception of the troubled spots
C. The troubled spots are more perceptible than the horizon
D. The trouble spot of the horizon have not been noted

48. The events of last Friday show that there have been no love lost between the principal and vice principal A. They dislike each other
B. They dislike one another
C. They work independently
D. They cannot part company
49. Ovie's account with the commercial bank is in red
A. Ovic's account was written red
B. Ovies account is in danger
C. Ovie's account is overdrawn
D. Ovie's account is special
50. People may not pick flowers in the park A. People can pick flower in the park
B. People are prohibited from picking flowers in the park
C. People may not wish to pick flowers from the park
D. People cannot pick flowers from the park
51. Evaluate $\int_0^{\pi} \cos x \, dx$ A.1 B. C.2 D. π
52. The geometric mean of 4 and 9
A. 13 B.5 C.36 D.6
53. The sides of a right angled triangle is xcm, (x+1)cm and (x+2)cm. Find x
A.5 B.6 C.3 D.9
54. Factorize $x^3 + 1$
A. $(x+1)^3$ B. $(x+1)(+1)(x+1)$
C. $3(x+1)$ D. $(x+1)(x^2-x+1)$
55. Give that $\log_5 2 = 0.431$ and $\log_5 3 = 0.682$, the value of $\log_5 \frac{3}{8} + 2\log_5 \frac{4}{5} - \log_5 \frac{2}{5}$ is
A. -1.271 B. -0.318
C. -2.104 D. -3.011
56. Evaluate $(625)^{-1/4}$ A. $3/11$ B. $1/7$
C. $1/5$ D. $9/10$
57. Find the universal set $U = \{x: -3 < x \leq 4\}$ and its subset $A = \{-1, 1, 2\}$ $B = \{x: 1 \leq x \leq 3\}$ Find $A \cap B$
A. $\{-1, 3\}$ B. $\{-3, -2, 1\}$
C. $\{-1, 0, 3, 4\}$ D. $\{1, 2\}$
58. The area of a triangle of sides 5cm, 6cm and 7cm is
A. $6\sqrt{6}\text{cm}^2$ B. 120cm^2
C. $2\sqrt{11}\text{cm}^2$ D. $4\sqrt{7}\text{cm}^2$
59. The perpendicular distance of a chord from the centre of a circle of radius 10cm is 6cm. Find the length of the cord.
A. 16cm B. 26cm C. 9cm D. 32cm
60. At what point does the curve $x^2 - 3x + 2$ cut the line $y = 0$
A. $x = 2, x = 1$ B. $x = -1, x = -2$
C. $x = -2, x = 1$ D. $x = 2, x = -1$
61. Simplify $2^{-3} \times \frac{163}{4}$
A. 8 B. 2 C. 4 D. 1
62. If x varies directly as vy and $x = \sqrt{9}$ when $y = 9$, then the value of x when $y = 17/9$ is
A. 9 B. 4 C. 16 D. 21
63. Ann walks for 5m from a point A to a point B Northwards. From B, she walks eastwards to a point C on the bearing 030° . How far is she from B?
A. $\frac{\sqrt{3}}{3m}$ B. $\frac{\sqrt{3}}{2m}$
C. $\frac{\sqrt{3}}{2m}$ D. $\sqrt{5}m$
64. A machine gun with a mass of 60kg fires a 0.2kg bullet at a speed of 200m/s. What is the recoiling speed of the machine gun?
A. 0.5m/s B. 1.0m/s
C. 1.5m/s D. 2.0m/s
65. Which of the following is an instrument for measuring temperature? A. Liquid in glass thermometer
B. Constant volume gas thermometer
C. Thermoelectric Thermometer
D. All of these
66. How far from a tall wall should a girl stand in order to hear the echo of her clap two second later (speed of sound in air equal 330m/s)?
A. 130cm B. 165cm
C. 210m D. 330m
68. Impulse is equal to____
A. Momentum
B. change in momentum
C. change in momentum per unit time
D. change in momentum per unit mass

69. An 800kg car moving at 80km/hr overtakes a 1200kg car moving at 40km/h in the same direction. If the two cars stick together, the wreckage has an initial speed of
A. 8km/h B. 40km/h
C. 56km/h D. 60km/h
70. The law which states that the volume of a given mass of gas increases linearly with temperature provided the pressure of the gas remain constant is referred to as
A. Boyle's law B. Charles's law
C. Pressure law D. Ideal law
71. A copper block has heat capacity of 68J/K. What is the mass of the block given that the specific heat capacity of copper is 400J/kgK
A. 0.07kg B. 0.17kg
C. 0.76kg D. 0.97kg
72. Two resistance joined in parallel produced a resultant of 6/5ohm. One of the resistance wires is broken and the effective resistance becomes 2ohm. The resistance of the wire that got broken is
A. 2 ohm B. 3 ohm
C. 3/5 ohm D. 0.6/5 ohm
73. An electrolytic capacitor has a capacitance of 0.005F. What is the charge on the plate if it is charged to a potential of 240 V.
A. 4.8C B. 3.6C
C. 2.4C D. 1.2C
74. Which of the following is NOT a scalar quantity?
A. Mass B. Volume
C. Weight D. Current
75. A car travels 10km due east on a straight road and then 15km due north before coming to rest. What is the resultant displacement of the car?
A. 5km B. 12.5km
C. 18km D. 25km
76. Which of the following metals will show the highest metallic character?
A. Calcium B. Barium
C. Copper D. Zinc
77. The oxidation state of Cr in $K_2Cr_2O_7$ is
A. +7 B. +6 C. +5 D. +4
78. The ores that can be concentrated by flotation are
A. nitride ores B. sulphide ores
D. Chloride ores D. Chloride ores
79. When temporary hard water is boiled for some time in a kettle, the inner surface of the kettle becomes coated with a deposit
A. Calcium tetroxosulphate (VI)
B. Calcium trioxocarbonate (IV)
C. Calcium hydrogentrioxocarbonate (IV)
D. aluminium trioxocarbonate (IV)
80. Which of the following is used to hasten the ripening of fruits?
A. Ethene B. Ethanol
C. Ethyne D. Ethane
81. The maximum number of electrons in the L-shell of an atom is
A. 2 B. 8 C. 18 D. 32
82. A solid substance with high melting and boiling points is likely to be a/an
A. covalent compound
B. dative covalent compound
C. electrovalent compound
D. Non-metal
83. The advantage of detergent over soap at detergents
A. are readily available
B. are in powdered form
C. are non-biodegradable
D. lather readily with water
84. Which of the following metals is purified commercially by electrolysis?
A. Zn B. Fe C. Sn D. Cu
85. The partial pressure of oxygen in a sample of air is 452mmHg and the total pressure is 780mmHg. What is the mole fraction of oxygen?
A. 0.203 B. 0.579
C. 2.030 D. 5.790
86. The component of air that can be removed by alkaline pyrogallol solution is
A. oxygen B. Nitrogen
C. carbon (IV) oxide D. noble gases
87. What quantity of aluminium is deposited when a current of 10A is passed through a solution of an aluminium salt for 1930s? (Al = 27, F = 96,500Cmol⁻¹).

- 
- World
Scholarship
Forum**

likely to have been obtained in a cupboard?

- A. 15sees B. 30sees
C. 25 sees D. 20 sees

SOLUTION

1. B 2. A 3. C 4. A 5. B 6. B 7. B
8. D 9. B 10. C 11. A 12. B 13. C 14. B
15. D 16. D 17. A 18. D 19. D 20. C 21. C
22. C 23. A 24. 25. A 26. C 27. A 28. B
29. C 30. D 31. D 32. B 33. D 34. A 35. C
36. A 37. A 38. D 39. B 40. A 41. A 42. D
43. C 44. D 45. A 46. B 47. A 48. A 49. C
50. C 51. A 52. D 53. C 54. D 55. B 56. C
57. D 58. A 59. A 60. A 61. D 62. B 63. A
64. C 65. D 66. D 67. A 68. B 69. C 70. B
71. B 72. B 73. D 74. C 75. C 76. A 77. B
78. B 79. B 80. A 81. B 82. C 83. D 84. D
85. B 86. A 87. B 88. C 89. D 90. C 91. D
92. C 93. D 94. D 95. C 96. 97. B 98. D
99. C 100. A

POST-JAMB SCREENING TEST
2013/2014
REVISION QUESTIONS AND ANSWERS

ENGLISH

In each of the question 1-5 choose the option nearest in meaning to the word(s) or phrase(s) in italics (underlined)

1. We were at a loss to understand if Edesiri should retract his statement before the judge
A. Assert B. Repeat
C. Withdraw D. Reiterate
2. Your effort to bring peace between us is futile because you are biased
A. Convinced B. Clever
C. Prejudiced D. Worried
3. This embroidery was an exacting task for a tailor
A. Easy B. Tasking
C. Interesting D. Beautiful
4. I had to make my way through the throng of people in the church
A. Group B. Crowd
C. Gathering D. Theme
5. You're impossible she said tome A.
Unreasonable B. Unforgiving
C. Daft D. Strong

Read the passage below and answer the questions that follows:

In times past in Nigeria, being a secretary was perceived with utter disdain by many people. Other professionals such as teachers, lawyers, engineers, surveyors, estate officers and architects were revered. Many issues readily come to mind concerning the old perspective of secretaries as professionals. One, they were regarded as people with little or no education. Indeed, many of them were said to be mere typist who learnt how to punch keys on the typewriters. They hardly had any formal education but obtained doubtful certificates. This were made possible because anyone could easily learn to be typist under the shade of a tree or in a personal office of a retired secretary or a chief typist from any business organization. As long as a short piece of work could be typed almost accurately and in record time then became a secretary. This unrealistic practice-to

call an unqualified person a secretary- might have been introduced and sustained by people who did not know who a secretary was. The educated secretaries obtained a minimum of the Ordinary National Diploma (OND) or its equivalent from institutions in and outside Nigeria particularly Great Britain; Canada or the United States of America. Two, they were mostly women and very few male secretaries who suffered more humiliation than their female counterparts. They were regarded as men who did not succeed in life and did the work that normally women were supposed to do. Three and deriving from the second reason; secretaries were, as a matter of fact, very close to their bosses. Most people, including husbands of secretaries regarded such a mode of interaction as tempting a kind of constant closeness which might engender a deliberate or unintentional amorous relationship between 'superior and subordinate officers.

6. Modern secretaries arc different from the old one because they:
A. Can use modern gadgets such as computer and other machines
B. Are able to become graduates of various universities
C. May become teachers in Polytechnics and Universities
D. Are more educated and better exposed to professional training
7. The expression amorous relationship as used in the passage means A. Relate as professional
B. Amateur relationship
C. Sexual relationship
D. Academic relationship
8. The expression utter disdain in the passage means A. Complete contempt
B. Deserved respect
C. Professional misconduct
D. Undue criticism
9. Which of the following statements can be deduced from the passage concerning the old perspective of secretaries?
A. They were highly respected by other professionals
B. They did not deserve to be addressed as secretaries
C. The training they received from retired professionals was sufficient
D. One could see that they were well educated.

10. The writer seems to suggest that:
 A. The Ordinary National Diploma was sufficient for secretaries
 B. Only graduates who studied abroad were good secretaries
 C. There was nothing wrong in having males as secretaries
 D. Secretaries were best trained by retired chief typists

In each of question 11 - 15 fill each gap with the most appropriate option from the list following the gap.

11. I was ____ hearing distance of the speaker
 A. On B. At C. Within D. In
12. My wife and I were to celebrate our silver wedding anniversary last Sunday.
 Unfortunately on that ____ day my father-in-law died mysteriously
 A. Fruitless B. Faithful
 C. Futile D. Fateful
13. The ceremony was rounded ____ very late
 A. Up B. Off C. Of D. Out
14. The ____ Affairs officer is expecting all of us in the dining room
 A. Student B. Students
 C. Student's D. Student's
15. My boss is usually a charming man but ____ he can be very irritable
 A. A times B. At times
 C. Those times D. With times

For each of questions 16-20, choose the option opposite in meaning to the word in italics (underlined)

16. Since mortality is the lot of man, argued the preacher, we must all accept death with equanimity
 A. Concern B. Uncertainty
 C. Serenity D. Anxiety
17. You can use lethal means if necessary to stop dangerous characters from molesting innocent citizens
 A. Fatal B. Cruel

C. Harmless D. Soft

18. The undesirable elements who perpetuate these vices are never detected
 A. Aggravate B. Overlook
 C. Discontinue D. Encourage
19. There is much indifference in his behaviour
 A. Concern B. Ignorant
 C. Unconcerned D. Inconsistency
20. The man found his wife physically repellant
 A. Attractive B. Fatuous
 C. Unpleasant D. Flamboyant

In question 21 - 25 choose the option nearest in meaning to the word(s) or phrase(s) in italic

21. Travelling in the night can be an ordeal
 A. A great exercise
 B. A great fear
 C. A great endurance
 D. A suffering
22. The confused student found himself in a spot
 A. In a particular space
 B. In a normal situation
 C. In a pleasant situation
 D. In a difficult situation
23. Our local paper is noted for its incisive editorials
 A. Sarcastic B. Penetrating
 C. Decisive D. Destructive
24. He is bound to expose himself by his nefarious activities
 A. Kind B. Disreputable
 C. Despicable D. Nocturnal
25. The Island became more enchanting at night
 A. Attractive B. Delightful
 C. Accommodation D. Lively

GENERAL PAPER

26. An inductance coil of 0.8 Henry is connected to a circuit with alternating A C source of 50 Hz. The reactance of the coil is
 A. 80 ohms B. 40 ohms
 C. 251.4 ohms D. 40×10^{-3} ohms
27. An appliance rated 220Vrms can with stand voltage values up to
 A. 440V B. 1000V
 C. 311V D. 660V

28. A transformer with 600 turns of coil in the primary arm and 30 turns of secondary coil carries a current of 0.65 A in the primary arm. The current in the secondary coil is
A. 0.033 Amp B. 20 Amp
C. 5Amp D. 13 Amp
29. Lightening conductor function effectively because
A. Charges settle at the tip of body
B. The atmosphere is big
C. It is connected to high tension source
D. it points to the North
31. A temperature of 78°C in Kelvin scale will be
A. 108.4k B. 351K
C. 140.4k D. 195k
31. How long will it take a body moving with an initial velocity of 15m/s to increase its velocity to 45m/s if it accelerates at 2m/s²?
A. 30secs B. 9secs
C. 2secs D. 15secs
32. Which of these best defines "resistivity" of a wire
A. Resistance per unit length
B. Resistance of a given area per unit length
C. Resistance of given length per unit area
D. Resistance per unit area
33. One horse power engine produces
A. 200watts B. 476watt
C. 1000watts D. 746watts-
34. Short sightedness is corrected by the use of
A. Diverging mirror B. Converging lens
C. Converging mirror D. Diverging lens
35. If a wave travelling with a velocity of 1200m/s has a frequency of 30Hz, its wavelength is
A. 3,600m B. 0.025m
C. 40m D. 1230m/s
36. A man borrowed NP at simple interest for 4 years at 3% and paid an interest of N240. What is the value of P?
A. 2000 B. 2400
C. 2880 D. 24000
37. Given that $\tan x = 1$ find $x + \cos x$
A. $2\sqrt{2}$ B. $\frac{1}{2}\sqrt{2}$ C. 2 D. $\sqrt{2}$
38. Write down the quadratic equation whose roots are P and -2P
A. $x^2 - px - 2p = 0$
B. $x^2 + px - 2p = 0$
C. $x^2 + px + 2p = 0$
D. $x^2 + 2px - p = 0$
39. Solve the equation $\frac{1}{\frac{1}{2}} = \frac{2}{\frac{1}{2+x}}$
A. $1\frac{1}{2}$ B. $-1\frac{1}{2}$ C. $\frac{1}{5}$ D. $\frac{3}{7}$
40. If a number is selected at random from the set what is the probability that it is prime?
A. $\frac{1}{6}$ B. $\frac{4}{7}$ C. $\frac{2}{3}$ D. $\frac{1}{2}$
41. The reactance of a capacitor whose capacitance is $8\mu f$ a circuit with 50Hz alternating effect is
A. 398Ω B. 2982Ω
C. 1982Ω D. 2202Ω
42. The angle between the plane mirrors in a kaleidoscope is usually
A. 90° B. 45° C. 60° D. 0°
43. Convert 42 base 5 to a number in base 3
A. 222 base 3 B. 211 base 3
C. 210 base 3 D. 112 base 3
44. A boy spends $\frac{1}{3}$ of his pocket money on books and $\frac{1}{4}$ on a pair of shorts. What fraction of his pocket money is left?
A. $\frac{2}{5}$ B. $\frac{5}{12}$
C. $\frac{5}{6}$ D. $\frac{1}{12}$
45. Simplify
A. 8 B. 4 C. 6 D. 7
46. The percentage of water of crystallization in $ZnSO_4 \cdot 7H_2O$ is ____ [Zn = 65, S = 32, O = 16, H = 1]
A. 33% B. 44%
C. 55% D. 87%
47. The soft solder used for welding and plumbing is an alloy of
A. Lead and zinc B. Iron and zinc
C. Iron and tin D. Lead and tin

48. 16.8g of sodium hydrogen trioxocarbonate (IV) is completely decomposed by heat. Calculate the volume of carbon(IV)oxide given off at s.t.p.
 A. 22.40dm^3 B. 11.20dm^3
 C. 2.24dm^3 D. 1.12dm^3
 (Na= 23, C = 12, O = 16, H = 1 molar volume of a gas at s.t.p. = 22.4dm^3)
49. Nigerian crude oil is described as light crude because of its
 A. Low sulphur content
 B. High natural gas content
 C. High octane number
 D. Low aliphatic hydrocarbon content
50. Which of the following acts as both a reducing and an oxidizing agent?
 A. H_2S B. CO_2 C. CH_2 D. SO_2
51. If and find the value of
 A. 14 B. 21 C. 32 D. 39
52. In a group of 30 students N of them are boys. If a student is selected at random, what is the probability that it is a girl?
 A. $\frac{30-n}{30}$ B. $\frac{10+n}{30}$
 C. $\frac{n}{30}$ D. $\frac{20}{30}$
53. Find the value of y if
 A. 7 B. 12 C. 9 D. 2
54. Find the value of the definite integral
 A. 4 B. 18 C. 0 D. 9
55. Each exterior angle of a regular polygon is 72° .
 A. 14 B. 9 C. 7 D. 5
56. The component of an atom that contributes least to its mass is the
 A. Nucleus B. Neutron
 C. Electron D. Proton
57. The soil around a battery manufacturing factory is likely to contain a high
 A. Ca^{2+} salts B. Pb^{2+} salts
 C. Mg^{2+} salts D. Al^{3+} ions
58. Haematite is an ore of
 A. Zinc B. Lead
 C. Iron D. Copper
59. Which of two structures are present in a palisade cell but not in a liver cell?
 A. Cell wall and cytoplasm
 B. Cell wall and chloroplast
 C. Cell membrane and cytoplasm
 D. Cell membrane and chloroplast
60. The hepatic portal vein in most cases is rich in food nutrients because it carries blood
 A. Directly from the heart
 B. From the liver to the heart
 C. From the small intestine to the liver
 D. From the liver to join the posterior vena cava
61. If a solution contains 4.9g of tetraoxosulphate(vi) acid, calculate the amount of copper (II) oxide that will react with it.
 (Cu= 64, O= 16, S=32, H= 1)
 A. 40.0g B. 80.0g
 C. 0.8g D. 4.0g
62. What mass of K_2CrO_4 is required to prepare 250cm^3 of 0.020mol dm^3 solution.
 A. 97.10g B. 19.42g
 C. 9.70g D. 0.97g
 [$\text{K}_2\text{CrO}_4 = 194.2\text{gmol}^{-1}$]
63. Hydrogen can be displaced from a hot alkaline solution by
 A. Sn B. Fe C. Cu D. Ca
64. The gas that can best be collected by downward displacement of air is
 A. Chlorine
 B. Sulphur (iv) oxide
 C. Carbon(iv)oxide D. Ammonia
65. Which of the following represents hybridization in ethyne?
 A. sp B. sp^3 C. sp^2 D. sp^2
66. If you cross-breed a tall variety (Tt) of maize with a short variety (tt), the ratio of the tall to short plants in the offspring will be
 A. 2:1 B. 1:2 C. 1:1 D. 1:3

67. Oxygen from the atmosphere reaches the mammalian bloodstream through one of the following pathways indicated below a- mouth b- trachea c-bronchiole d-nostril e-bronchus f-alveoli
A. A'!b'!e'!f'! D. B'!d'!f!e!e
C. D'!b'!e'!e'!f D. D'!'!f!e'e
68. When a person with blood group O loses a lot of blood due to accident, the person can be given blood from person of blood group
A. O only B. A only
C. AB only D. A and O
69. Which of the following statement about the rate of transpiration is incorrect? It is
A. Dependent on temperature
B. Affected by changes in light intensity
C. Unaffected by humidity
D. Affected by availability of water
70. Which of the following combination differentiates a bony fish from a cartilaginous fish?
I-presence of gills II - absence of gill
III-possession of a bony skeleton
IV - possession of laterally compressed body V-possession of dorso - ventrally compressed body
A. I, II and III B. I, III and IV
C. II, III and IV D. II, III and V
71. Which of the following occurs when a person looks at a distant, dim object and then look at a much closer bright objects? In the eve, A. Larger, while the lens becomes thicker
B. Larger, while the lens becomes thinner
C. Small, whole the lens becomes thicker
D. Smaller while the lens becomes thinner except
72. The role of ATP includes all the following except
A. Provisionary energy to do work in living organism
B. Transmission of nerve impulse
C. Keeping of body warm in homothermic animals
D. Producing water during aerobic respiration
73. Algae bloom depletes oxygen content of water
A. Industrial effluent
B. Thermal plant discharge
C. Raw sewage disposal
D. Crude oil spillage
74. Hypogeal germination is characterized by the
A. Emergence of the plumule, out of the ground
B. Provision of nourishment by the endosperm
C. Elongation of the hypocotyls
D. Elongation of the epicotyls.
75. Which one of the following statement about the mammalian kidney is false
A. The glomerulus to formed by a branch of the renal artery
B. The glomerulus is a part of the uriniferous tubules
C. The Bowman's capsule is in the center of the kidney
D. The uriniferous tubules are richly supplied with blood vessels.

Apt. Test

76. In the SUG election, the Republican candidate received one and a half times as many votes as the Democrat candidate. The Democrat candidates received one third more votes than the independent candidate 900 votes were cast for the independent candidates. How many votes were cast for the Republican candidates?
A. 900 B. 1,400 C. 1,600 D. 1,800
77. Anna and John both drive to their new home 400 miles away. Anna drives the family car at an average speed of 60 mph. John drives the removal truck at an average speed of 50 mph. During the journey, Anna stops for a total of 1 hour and 20 minutes, John stops for a half a long. What is the difference in minutes between their arrival times?
A. 60 B. 55 C. 40 D. 90
78. Pedro goes either hunting for fishing every day. If it is snowing and windy then Pedro goes hunting. if it is Sunny and not windy then Pedro goes fishing. Sometimes it can be snowing and sunny Which of the following statements must be true?
A. If it is not sunny and it is snowing then Pedro goes hunting
B. If it is windy and Pedro does not go hunting then it is not snowing

- C. If it is windy and not sunny then Pedro goes hunting
D. If it is windy and sunny then Pedro goes hunting
79. A team of eight lumberjacks cut an average of 15,000 cubic feet of timber in a week. How many cubic feet will four lumberjacks cut in four weeks?
A. 30,000 B. 25,000
C. 32,000 D. 16,000
80. A rectangular solid is 19 inches high, 19 inches wide and 19 inches long. What is its volume in cubic inches?
A. 7,000 B. 7,500
C. 8,000 D. 8,500
81. Identify the missing number

A. 120 B. 122 C. 130 D. 132
82. It costs a manufacturer x naira per component to make the first 1,000 components. All subsequent components cost $x \div 3$ each. When x N1.50, how much will it cost to manufacture 4,000 components?
A. N3,500 B. N3,000
C. N4,000 D. N3,250
83. Sabrina's boss states that she will increase Sabrina's salary from N12,000 to N14,000 per year if she enrolls in business courses at a local community college. What percent increase in salary will result from Sabrina taking the business courses? A. 16.67% B. 26% C. 40% D. 10%
84. If all the 6 arc replaced by 9, then the algebraic sum of all the numbers from 1 to 100 (both inclusive) varies by
A. 330 B. 324
C. 1200 D. 412
85. From a pack of cards Jack, Queen, King & Ace arc removed. Then the algebraic sum of rest of the cards is
A. 217 B. 218
C. 216 D. 213
86. Swetha, Tina, Uma and Vidya are playing a gambling, in this different people lose in

different games- in the reverse alphabetical order the rule is the if one loses she should double the among of others. At the end of 4th game each of them have same amount of money (N32.00) Which one of them started with the least amount? A. Vidya B. Tina
C. Uma D. Swetha

87. In a college, some courses contribute more towards an overall GPA than other courses. For example, a science class is worth 4 points; mathematics is worth 3 points; History is worth 2 points; and English is worth 3 points. The values of the grade letters are as follows, A 4, B3,C2,D1,F=0. What is the GPA of a student who made a "C" in the Trigonometry, a "B" in American History, an "A" in Botany, and a "B" in Microbiology?
A. 2.59 B. 2.86 C. 3.08 D.3.33
88. If $8x+5x+2x+4x=114$, then $5x+3=$ A. 12 B. 25
C. 33 D. 47
89. During a 5 - day festival, the number of visitors tripled each day. If the festival opened on a Thursday with 345 visitors, what was the attendance on that Sunday?
A. 345 B. 1725
C. 3,196 D. 9315

Please select the answer choice that identifies the noun in the sentence in 90-92

90. It will take all of your energy and will be able to walk again
A. Take B. All
C. Yours D. Energy
91. Sharks and Lampreys are not true fish because their skeletons are made of cartilage rather than bone
A. True B. Because
C. There D. Bone
92. The Trojans' rash decision, to accept the wooden horse, led to their destruction
A. Their B. Led
C. Accept D. Destruction
93. In Microsoft word, Times New Roman is the name of
A. A font in MS Office B. Paragraphing
C. Escape button D. Just Roman time

94. Sentence is to paragraphs as word is to
A. Letter B. Sentence
C. Paragraph D. Phrase
95. Which of the odd one out A. Marriage B. Annul
C. Divorce D. Separation
96. Sarah needs to make a cake and some cookies. The cake requires $\frac{3}{8}$ cup of sugar and the cookies requires $\frac{3}{5}$ cup of sugar. Sarah has $\frac{15}{16}$ cups of sugar. Does she have enough sugar, or how much more does she need? A. She has enough sugar
B. She needs $\frac{1}{8}$ of cup of sugar
C. She needs $\frac{3}{80}$ of a cup of sugar
D. She needs $\frac{4}{19}$ of a cup of sugar
97. If Judy comes to the party then Sally leaves the party. If Sally leaves then either Christine or Clara ask Philip to dance. If Philip is asked to dance by either Christine or Clara and Sally leaves the party, Philip accepts. If Philip is asked to dance by either Christine or Clara and Sally does not leave the part Philip does not accept. If sally does not leave the party, which of the
A. True B. Because
C. There D. Bone

Correct the error in each underline word, phrase or sentence in 98 and 99

98. Jessica Mitford wrote the American way of Death, a best-seeling book that led eventually to an official investigation of the funeral industry.
A. That led eventually
B. That had led eventually
C. That eventually led
D. Which led eventually
99. Once upon a point in time, a small person named Little Red Riding Hood initiated plans for the preparation, delivery and transportation of foodstuffs to her Grandmother. A. And transportation of foodstuffs to her Grandmother
B. And transportation of food stuffs to her grandmother
C. And transportation of food supplies to her grandmother

D. And transportation of foodstuffs to her grandmother

100. They key on the keyboard that is used to capitalize letter when pressed with that letter key is
A. Shift B. Hold
C. Caps lock D. Enter

SOLUTION

- | | | | | | | |
|-------|--------|-------|-------|-------|-------|-------|
| 1. C | 2. C | 3. B | 4. B | 5. A | 6. D | 7. C |
| 8. A | 9. B | 10. C | 11. C | 12. D | 13. B | 14. A |
| 15. B | 16. D | 17. D | 18. C | 19. A | 20. A | 21. C |
| 22. D | 23. B | 24. C | 25. B | 26. B | 27. C | 28. D |
| 29. A | 30. B | 31. D | 32. B | 33. D | 34. D | 35. C |
| 36. D | 37. D | 38. B | 39. | 40. A | 41. A | 42. C |
| 43. B | 44. D | 45. | 46. B | 47. D | 48. C | 49. C |
| 50. D | 51. | 52. A | 53. | 54. | 55. D | 56. C |
| 57. B | 58. C | 59. D | 60. A | 61. D | 62. D | 63. D |
| 64. D | 65. A | 66. D | 67. C | 68. A | 69. C | 70. B |
| 71. B | 72. B | 73. C | 74. A | 75. C | 76. D | 77. B |
| 78. B | 79. A | 80. A | 81. B | 82. B | 83. A | 84. C |
| 85. | 86. D | 87. B | 88. C | 89. D | 90. D | 91. D |
| 92. D | 93. A | 94. D | 95. D | 96. | 97. C | 98. C |
| 99. D | 100. C | | | | | |

POST-JAMB SCREENING TEST REVISION QUESTIONS AND ANSWERS 2012/2013 SESSION DELSU AFTERNOON SESSION

ENGLISH

Read the passage below and answer the question that follow:

You should think that the common cold should be to easy enough to study, but it is not so easy as it looks. Colds often seen to spread from one person to another, so it is often assumed that the cold must be infectious, but there are some puzzling observations which do not fit in with this theory. An investigator in Holland examined some eight thousand volunteers from different areas and came to conclusion that in each group the colds all appeared at the same time – transfer of infection from case to case could not account for that. Yet at the common cold research institute in Salisbury the infection theory has been tested out, two series of about two hundred people each were inoculated one with salt water and the other with secretions from known cold victims. Only one of the salt water group got a cold, compared with seventy three in the other group.

In the British Medical Journal, there was a report of a meeting "The common cold-fact and fancy" at which one of the speakers reported a study of colds made in Cireneester over the last five years. Three hundred and fifty volunteers had kept diary records of their colds and on an average each had seven every year, with an annual morbidity of seventy days. So, nearly one-fifth of our lives is spent in more or less misery coughing and sneezing. Some widely held beliefs about the common cold have turned out not to be true. It seems that cold people are just as liable to colds as the young. Sailors in isolated weather ships have just as colds while on board and not in contact with the outside world as when on shore.

1. Which of the following statements can be implied from the passage
 - A. People catch colds equally in warm and cold weather
 - B. People catch more colds in water
 - C. People catch more colds in warm weather
 - D. The origin of cold is inconclusive
2. According to the writer, some widely held beliefs about the common colds are:
 - A. Inconclusive B. Inevitable
 - C. Fallacious D. Societal
3. The cirencester volunteers kept a record of their colds through
 - A. The British Medical Journal
 - B. Personal diaries
 - C. Morbidity rates
 - D. Temperature recordings
4. A rare disease can be more easily dealt with than the common cold because
 - A. Medical experts are fed up with the rampant cases of common cold
 - B. Common colds are easily not the province of the orthodox medical experts
 - C. People easily develop resistance to the common colds
 - D. Adequate research exists to uncover facts about such rare diseases
5. From the information in the passage, there is evidence
 - A. That old people are immune against the common cold
 - B. Against the theory that the common cold is infectious
 - C. For the theory that common cold is indeed infectious

- D. That medical reports are silent on facts about common colds

In questions 6 - 13 choose the option nearest in meaning to the word(s) or phrase(s) in italics (underlined)

6. All the accident victims are bleeding profusely
 - A. Thorough B. Perfectibility
 - C. Complete D. Practicability
7. The Prosecutor was accused of obstructing justice
8. The man preaches egalitarianism without matching it up with action
 - A. Salvation B. Equality
 - C. Kindness D. Dedication
9. It was surprising how we took to the stranger at once
 - A. Talked so long with
 - B. Introduced ourselves
 - C. Formed a liking for
 - D. Started fighting
10. Surely, Toritseju is a ball of fire
 - A. A quarrelsome person
 - B. A hot hearted activist
 - C. A fiery speaker or writer
 - D. A very energetic person
11. It would need a high flyer to make a first class degree in university
 - A. A smart performer
 - B. An outright genius
 - C. An outstanding scholar
 - D. An unmitigated swot
12. Hers was a chequered career
 - A. An interesting and successful career
 - B. A career full of sorrow and fears
 - C. A bright and memorable career
 - D. A career full of ups and downs

In each of questions 13 - 18, choose the option opposite in meaning to the word in italics (underlined)

13. The flock suddenly became restive after midday
 - A. Quiet B. Disorderly
 - C. Hady D. Fidgety

14. Edesiri's aberrant behavior attracted attention
A. Rascally B. Hady
C. Normal D. Selfish
15. The judge always takes a hardline on such issues
A. Complex stand B. Disobedient
C. Uncompromising D. Difficult stand
16. That team has become quite indomitable quite recently
A. Incurable B. Disobedient
C. Unconquerable D. Conquerable
17. Okoro is an amateur wrestler
A. Skillful B. Good
C. Professional D. Strong
18. The instructions on the examination paper and explicit
A. Simple B. Detailed
C. Ambiguous D. Definite

in each of the questions 19-22 choose the option nearest in meaning to the word(s) or phrase(s) in Italics (underlined)

19. The police ran the criminal to earth
A. Jailed
B. Knocked him down
C. Discovered him down
D. Buried him
20. The college was worried by Ejoro's garrulous tendencies
A. Reticent B. Loquacious
C. Quarrelsome D. Outspoken
21. The economic situation is responsible for the recent discord in many families
A. Division B. Resourcefulness
C. Harmony D. Suffering
22. The awareness of mortality has imposed some restraint on man's excesses
A. Celestially B. Eternity
C. Continuity D. Inexistence

In each of questions 23 - 25 fill each gap with the most appropriate option from the list following the gap.

23. You think she hasn't got any friends? of course she's got _____
A. Many more B. Quite much
C. Quite a few D. Few
24. The carpenter asked for ten packets of _____ nails
A. Fifteen-centimetres
B. Fifteen-centremetres
C. Fifteen-centimeter
D. Fifteen-centemeters
25. My brother got married to a women with _____ manner
A. Gorgeous B. Exquisite
C. Decent D. Attractive

Gen. Paper

26. The type of bonding in $(\text{Cu}(\text{NH}_3)_4)^2$ is
A. Coordinate B. Electrovalent
C. Metallic D. Covalent
27. Which of the following represents an aromatic compound
A. C_6H_{12} B. C_6H_8
C. $\text{C}_6\text{H}_5\text{OH}$ D. $\text{C}_6\text{H}_9\text{OH}$
28. 300cm^3 of a gas has a pressure of 800mmHg. If the pressure is reduced to 640mmHg. Find its volume.
A. 243.75cm^3 B. 369.23cm^3
C. 738.46cm^3 D. 1733.36cm^3
29. A dense white fume is formed when ammonia gas reacts with
A. $\text{H}_2\text{S(g)}$ B. $\text{O}_2\text{(g)}$ C. HCl(g) D. $\text{Cl}_2\text{(g)}$
30. The enzyme responsible for converting sucrose into two simple isomeric sugar is
A. Invertase B. Amylase
C. Lactase D. Maltase
31. Oxyacetylene flame is used for iron-welding because it
A. Evolves a lot heat when burnt
B. Dissociate to produce carbon(iv)oxide and oxygen
C. Makes the iron metal solidify very quickly
D. Combines with oxygen to give a pop sound

32. The structural component that makes detergent dissolve more quickly in water than soap is
A. $-\text{SO Na}^+$ B. $-\text{COOK}^+$
C. $-\text{COONa}^+$ D. $-\text{SO}_3\text{Na}^+$
33. According to the kinetic theory an increase in temperature cause the kinetic energy of particles to
34. Chlorine gas is prepared in the laboratory by
A. Dropping concentrated hydrochloric acid onto potassium tetraoxomanganate(vii) crystals
B. Oxidizing concentrated hydrochloric acid using potassium heptaoxodichromate(vi) crystals
C. Adding concentrated tetraoxosulphate(vi) acid in solid sodium chloride
D. Adding concentrated hydrochloric acid to solid manganese(iv)oxide
35. A compound contains 40.0% carbon, 6.7% hydrogen and 53.3% oxygen if the molar mass of the compound is 180. Find the molecular formula. [H=1, C=12, O=16]
A. $\text{C}_6\text{H}_6\text{O}_3$ B. $\text{C}_6\text{H}_{12}\text{O}_6$
C. $\text{C}_3\text{H}_6\text{O}_3$ D. CH_2O
36. Pure solvents are obtained by
A. Evaporation
B. Extraction
C. Condensation D. Distillation
37. A given volume of methane diffuses in 20s. How long will it take the same volume of sulphur(iv)oxide to diffuse under the same conditions? [C = 12, H=1, S=32, O=16]
A. 40s B. 60s
C. 20s D. 5s
38. 25cm^3 of a 0.2mol dm^{-3} solution of Na_2CO_3 requires 20cm^3 of a solution of H^+Cl^- for neutralization. The concentration of the HCl solution.
A. 0.5mol dm^{-3} B. 0.6mol dm^{-3}
C. 0.2mol dm^{-3} D. 0.4mol dm^{-3}
39. Multiply 0.0012 by 0.0408 leaving your answer in standard form.
A. 4.896×10^{-3} B. 4.896×10^6
C. 4.896×10^{-5} D. 4.896×10^{-5}
40. The next two terms of the Arithmetic progression 212, 201, 190, 179, __, __, are
A. 168, 157 B. 168, 156
C. 157, 146 D. 175, 164
41. Simplify $\log_2 8 - \log_2 14 - \log_2 7$.
A. $\log_2 15$ B. $\log_2 4$
C. $\log_2 4$ D. 1
42. If x varies directly as y^2 and $x = 4$ when $y = 6$ find the value of y when x
A. ± 36 B. ± 9 C. $\frac{1}{9}$ D. 81
43. Determine for what range of values of x the inequality $\frac{1}{2}x^2 \geq 2x - 1$ is satisfied.
A. $x \geq 2$ B. $x \leq 2$
C. $x \leq 5$ D. $x \geq 5$
44. If $\frac{dy}{dx} = e^x + \sin x + 2x$ find y
A. $e^x + \cos x + x^2 + c$ B. $e^x + \cos x - x^2 + c$
C. $e^x + \cos x + x^2 + c$ D. $e^x - \cos x - 2x^2 + c$
45. A regular polygon has 8 sides. Find the size of its interior angles.
A. 45° B. 90° C. 135° D. 1080°
46. Factorise $75p^2x^2 - 48y^2$
A. $3(5px - 4y)(5px \pm 4y)$
B. $3(5px - 4y)(5px - 4y)$
C. $9(5px - 4y)^2$
D. $3(5px + 4y)^2$
47. The sum of the roots of the equation is
A. -18
B. -6 C. -4 D.
48. Given the universal set $U = \{1, 2, 3, \dots, 10\}$ set
 $P = \{x: x \text{ is odd}\}$ and
 $Q = \{x: x \text{ is prime}\}$ list the elements of $P \cap Q$
A. $\{3, 5, 7\}$ B. $\{1, 9\}$
C. $\{1, 3, 9\}$ D. $\{2, 4, 6, 8\}$
49. P is a point on the bearing 342° from a point Q. What is the bearing of Q from P?
A. 162° B. 198°
C. 252° D. 342°
50. Which of the following represents an inverse variation?
A. $y = 2x^2$ B. $y = \frac{1}{2}x$

C. $y = -4x$ D. $y = \frac{4}{x}$

51. Two of the interior angles of a polygon are 156° and 134° and each of the remaining interior angles is 151° . How many sides has the polygon
A. 5 B. 10 C. 12 D. 16
52. A temperature of 50°C is Fahrenheit scale.
A. 60°F B. 122°F
C. 28°F D. 90°F
53. The vacuum in the thermos flask prevent heat losses by
A. Conduction and convection
B. Radiation only
C. Convection and radiation
D. Conduction only'
54. Anomalous expansion of water occurs at.
A. 0°C B. 4°C C. 10°C D. 100°C
55. A small object is placed 60cm in front of a diverging (Concave) lens of focal length 20cm. Calculate the position of the image formed and what is the nature of the image? A. 15cm and virtual B. 30cm and real
C. 15cm and real D. 30cm and virtual
56. If a wave travelling with a velocity of 1200m/s has a frequency of 30Hz, its wavelength is
A. 3,600m B. 0.025m
C. 40m D. 1230m/s
57. Two plane mirrors inclined at 30° will produce image
A. 3 B. 5 C. 11 D. 12
58. A ray makes an angle of incidence of 30° as it travels from air to glass of refractive index 1.5. The angle of refraction is
A. 19.5° B. 48.6°
C. 45° D. 50°
59. A 20N force moves through a displacement of 300mm in 10 minutes. Its power is
A. 600watts B. 10watts
C. 1500watts D. 150watts
60. Which of these has the greatest penetrating power
A. Gamma B. X-rays
C. Beta D. Alpha
61. Which of these is/are rechargeable?
(i) Cadmium indium cell
(ii) Leclanche cell
(iii) Lead acid accumulator
A. i and ii B. ii and iii
C. ii only D. i and iii
62. The energy stored in an inductor of 5mH when a current of 6A flows through it is
A. $1.8 \times 10^2\text{J}$ B. $9.0 \times 10^{-2}\text{J}$
C. $9.0 \times 10^{-3}\text{J}$ D. $1.4 \times 10^{-2}\text{J}$
63. The sum of all currents flowing into a junction is equal to the sum of the currents flowing out. This is
A. Ohms Law B. Faraday's Law
C. Kirchhoff's Law D. Bragg's Law
64. Continuity of an ecosystem depends on constant flow of energy and circulation of matter. However, the amount of energy passed on from decomposers to producers should be
A. Great B. Small
C. Reversible D. Zero
65. A stable self sustaining environment produced by an interaction between biotic and abiotic component is best described
A. A niche B. A community
C. An ecosystem D. A habitat
66. A density dependent factor that regulate a population size may include the following
A. Shortage of food supply
B. Fire outbreak
C. Spread of disease
D. Increase competition
67. In the life history of Schistosoma (Bilharzia) one of the following is the intermediate host
A. Man B. Snail
C. Mosquito D. Crayfish
68. A human gland secretes two hormones, one of which causes a disease called diabetes if deficient in the body. Which organ procures the hormone

- A. Spleen
C. Adrenal body
- B. Parathyroid
D. Pancreas

69. A married couple have 10 children and they are all girls. Which of the following is the correct explanation?

- A. The woman is incapable of producing male children
B. The man's sperms are very weak
C. The x components of the man's sex chromosome are always involved
D. The male is not athletic enough.

70. If a 26 old blind man married a young one-eyed woman and they had four children, how many of them would be blind like their father?

- A. All B. 3 C. 2 D. None

71. A mixture of mercurous and mercuric nitrate is added to a food substance and a white precipitate is formed and when heated gently turns red. The food substance is

- A. Protein B. Oil
C. Carbohydrate D. Fat

72. Starting from one skull end, the vertebrates are arranged in the following order

- A. Axis, atlas, cervical, thoracic and lumbar
B. Atlas, cervical, axis, thoracic and lumbar
C. Atlas, axis, thoracic, cervical and lumbar
D. Atlas, axis, cervical, thoracic, and lumbar

73. Which of the following represents the sequence of protein hydrolysis? 1 - polypeptide 2 - Amino acids 3 - protein 4 - peptides

- A. 3'1'!2'!4 B. 3'!2'!4'!1
C. 3'!4'!2'!1 D. 3'!4'!1'!2

74. Which is the most important adaptation of a bony fish to life in water? A. the possession of a stream lined shape

- B. the presence of overlapping scales
C. the covering of the body by a film of slime
D. the possession of a caudal fin

75. Double fertilization in higher plants is significant because it ensures that

- A. formation of a fertile embryo
B. formation of a fertile embryo and the endosperm
C. development of the seed

D. development of the fruit.

76. A train travelling at 60 mph enters a tunnel that is 5 miles long. The train is one mile long. How many minutes does it take for the whole train to pass through the tunnel?

- A. 7 B. 4 C. 10 D. 5

77. The cost of manufacturing tractors in Ghana is twenty percent less than the cost of manufacturing tractors in Nigeria. Even after transportation fees and import taxes are added, it is still cheaper to import tractors from Korea to Nigeria than to produce tractors in Nigeria. Which of the following assertions is best supported by this information?

- A. Labour costs in Ghana are twenty percent below those in Nigeria
B. Importing tractors into Nigeria will eliminate twenty percent of the manufacturing jobs in Nigeria
C. The costs of transporting a tractor from Ghana to Nigeria is more than twenty percent of the cost of manufacturing the tractor in Ghana
D. The import taxes on a tractor imported from Ghana to Nigeria is less than twenty percent of the cost of manufacturing the tractor in Nigeria.

78. There are two valves at the bottom of a water tank which allow the tank to be drained. If both valves are opened, how long will it take before the tank is empty

Statement 1 - if only valve 1 is opened, the tank will be empty in 10 minutes
Statement 2 - if only valve 2 is opened, the tank will be empty in 10 minutes.

- A. Statement 1 alone is sufficient, but statement 2 alone is not sufficient
B. Statement 2 alone is sufficient, but statement 1 alone is not sufficient
C. Both statements together are sufficient, but neither statement alone is sufficient
D. Each statement alone is sufficient

79. Of the teams competing in the world archery championships, twenty percent are from Europe. Half as many are from the United States and one twentieth are from Africa. What fraction of teams are from neither Europe, the US nor Africa.

Identify the relationship between the word group in the question. Shade the answer which fits best with the group. 80. Throw, volley, string A. Hurl B. Grab

C. Seize D. Catch

81. Bourbon, whisky, gin
A. Bear B. Vodka
C. Linen D. Fur

Correct the error in each underlined word, phrase or sentence in 82 and 83.

82. Linda decides they had better scam before the killers find them. A. Had better scam
B. Had better leave
C. Should hurry and scam
D. Could hurry and leave

83. Quebee rises in a magnificent way above the St. Lawrence River
A. Rises in a magnificent way above
B. Rises magnificent way, way above
C. Rises magnificently above D. Rises magnificent way above.

84. If one-seventh of a number exceeds its eleventh part by 100 then the number is..
A. 770 B. 1100
C. 1825 D. 1925

85. A man can row 5 kmph in still water. If the river is running at 1kmph, it takes him 75 minutes to row to a place and back. How far is the place?
A. 3km B. 2.5km
C. 4km D. 5km

86. Swetha, Tina, Uma and Vidya are playing a gambling. In this different people lose in different games-in the reverse alphabetical order. The rule is that if one losses she should double the amount of others. At the end of 4th game each of them have same amount of money (N32). Which one of them started with the largest amount of money?
A. Vidya B. Tina
C. Uma D. Swetha

87. There are two circle, one circle is inscribed and another circle is circumscribed over a square. What is the ratio of area of inner to outer circle?
A. 1:2 B. 3:4 C. 2:1 D. 5:6

88. Mr. and Mrs. Aye and Mr. and Mrs. Bee competed in a chess tournament. Of the three games played.

- (i) in only the first game were the two players married to each other
(ii) the men won two games and the women won one game
(iii) the Ayes won more games than the Bees
(iv) anyone who lost a game did not play the subsequent game
Who did not lose a game?
A. Mrs. Bee B. Mr. Bee
C. Mr. Aye D. Mrs. Aye

89. Three piles of chips-pile 1 consists one chip, pile II consists of chips, and pile III consists of three chips-are to be used in game played by Anita and Brinda. The game requires:

- (i) that each player in turn take only one chip or all chips from just one pile
(ii) that the player who has to take the last chip loses
(iii) that Anita now have her turn From which pile should Anita draw in order to win

- A. Pile I B. Pile II
C. Pile D. None

90. In a class composed of x girls and y boys what part of the class is composed of girls?

- A. $\frac{y}{(x-y)}$ B. $\frac{y}{xy} \frac{x}{(x-y)}$
C. $\frac{x}{(xy)}$ D.

91. If the operation, is defined by the equation $xy = 2x+y$, what is the value of a in $2a = a3$
A.0 B.1 C. -1 D.4

92. There are 200 questions on a 3 hr examination. Among these questions are 50 mathematics problems. It is suggested that twice as much time be spent on each maths problem as for each other question. How many minutes should be spent on mathematics problems'? A. 36 B. 72 C. 60 D. 100

93. In a group of 15, 7 have studied Latin, 8 have studied Greek, and 3 have not studied either. How many of these studied both Latin and Greek
A.0 B. 3 C.4 D. 5

94. If a and b are positive integers and $(A-b)/3.5$ is $4/7$, then
A. $b < a$ B. $b > a$ C. $b = a$ D. $b > a$
95. A company contracts to paint 3 houses. Mr. Brown can paint a house in 6 days while Mr. Black would take 8 days and Mr. Blue 12 days. After 8 days Mr. Brown goes on vacation and Mr. Black begins to work for a period of 6 days. How many days will it take Mr. Blue to complete the contract?
A. 7 B. 8 C. 11 D. 12
96. Which of the following numbers could be described in the following way an integer that is a natural, rational and whole number?
A. 0 B. 1 C. 2.33 D. -3
97. In the fraction $3/x$, x may not be substituted by which of the following sets?
A. $\{1, 2, 4\}$ B. $\{-2, -3, -4\}$
C. $\{1, 3, 7\}$ D. $\{0, 10, 20\}$
98. Jim has 5 pieces of string. He needs to choose the piece that will be enable him to go around his 36-inch waist. His belt broke, and his pants are falling down. The piece needs to be at least 4 inches longer than his waist so he can tie a knot in it, but it cannot be more than 6 inches longer so that the ends will not show from under his shirt. Which of the following pieces of string will work the best?
99. The last week of a month a car dealership sold 12 cars. A new sales promotion came out the first week of the next month and the sold 19 cars that week. What was the percent increase in sales from the last week of the previous month compared to the first week of the next month?
A. 58% B. 119% C. 158% D. 175%
100. Select the answer choice that identifies the noun in the sentence. The works of many great poets have been placed on reserve
A. Many B. Great
C. Placed D. Reserve

SOLUTION

1. D 2. B 3. B 4. D 5. A 6. A 7. A
8. B 9. C 10. C 11. C 12. D 13. A 14. C
15. B 16. D 17. C 18. C 19. C 20. B 21. A

22. D 23. C 24. A 25. C 26. A 27. B 28. B
29. C 30. A 31. A 32. D 33. C 34. D 35. B
36. D 37. A 38. A 39. C 40. A 41. C 42. B
43. B 44. C 45. D 46. A 47. C 48. A 49. B
50. D 51. A 52. B 53. A 54. B 55. B 56. C
57. C 58. A 59. B 60. A 61. D 62. B 63. C
64. D 65. D 66. B 67. B 68. D 69. C 70. D
71. C 72. C 73. D 74. D 75. D 76. C 77. A
78. C 79. D 80. A 81. B 82. B 83. C 84. A
85. C 86. D 87. A 88. D 89. C 90. B 91. C 92. D 93. B
94. B 95. D 96. B 97. D 98. C 99. C 100. B

POST-JAMB SCREENING TEST REVISION QUESTIONS AND ANSWERS 2011/2012 SESSION DELSU

ENGLISH

Reading each passage and answer questions 1-5 Choosing the incorrect option

The main reason why the road leading from the main gate to lectures halls of campus III of Delta State University, Abraka is meandering is to regulate the speed of cars. So much landscaping is being done on the campus in order to improve the aesthetic beauty of the environment. Three faculty complexes viz, Education, Arts and science, are under construction and they will be completed before the end of the year. These constructions have, however, resulted in road diversions and other physical inconveniences on the campus. From the architectural design of the campus, it will be the most beautiful university campus in Nigerian when it is completed.

- The reason why the road leading from the main gate is windy is A. To increase aesthetic beauty
B. To ease flow of traffic
C. To regulate the speed of vehicles
D. To create diversions
- An appropriate title for the passage is A. Infrastructural development of campus III
B. The construction of three new faculty complexes
C. Beautification of campus III
D. Diversions on campus III
- From the passage we can gather that

- A. The inconvenience suffered by the inhabitants will be for a while
 - B. Projects are carried out without approval
 - C. There is deliberate effort to inconvenience the people
 - D. Buildings are put up anyhow
4. From the architectural design of the Campus in the passage means
- A. The structures on campus
 - B. The design of the three faculty complexes
 - C. The design of the road complex
 - D. The structural design of buildings on campus
5. The essence of landscaping is to
- A. Increase road network
 - B. Erect more buildings
 - C. Beautify the campus
 - D. Divert road network

Choose the option that best conveys the meaning of the underlined portion of each of the following sentences.

6. His jail terms were to run concurrently
- A. Consecutively
 - B. Uniformly
 - C. Simultaneously
 - D. Judiciously
7. I have an aversion for war films
- A. Preference
 - B. Dislike
 - C. Criticism
 - D. Bitterness
8. The teacher refused to answer for the mistakes made by the principal
- A. To give an answer
 - B. To explain
 - C. To accept responsibility for
 - D. To reply
9. When a man is immunized to an illness, he is
- A. Attached to it
 - B. Opposed to it
 - C. Protected against it
 - D. Addicted to it
10. In some parts of Nigeria, people are ostracized purely on the basis of their parentage.
- A. Disrespected
 - B. Hated
 - C. Abandoned
 - D. Shut off from society
11. Our town is so small that we are always meeting school mates accidentally
- A. Running into
 - B. Falling into
 - C. Coming across
 - D. Running up

12. To most people, last Christmas was an austere period
- A. Demanding
 - B. Irregular
 - C. Stimulating
 - D. Related to hercules
14. The witness was guilty of swearing to a statement he knew to be false
- A. Perjury
 - B. Slander
 - C. Forgery
 - D. Libel

Complete each of the following sentences by choosing the option that most suitably fills the space

15. Our plane was scheduled to _____ at 10.00hrs but it was delayed because of bad weather
- A. Takeoff
 - B. Fly out
 - C. Start out
 - D. Shoot off
16. As he was still owing me two hundred naira, I was careful not to _____ any more money
- A. Loan
 - B. Borrow
 - C. Extend
 - D. Lend
17. Remember this matter is strictly _____
- A. Between you and I
 - B. Among you and me
 - C. Between you and me
 - D. Between you and myself
18. The train railed along _____ full speed
- A. At
 - B. In
 - C. By
 - D. On
19. My niece is suffering _____ jaundice
- A. Of
 - B. For
 - C. With
 - D. From
20. Must you say "Good Morning" to everybody in the street? No _____
- A. I can't
 - B. I mustn't
 - C. I mustn't
 - D. I do not need

Choose the word that is opposite in meaning to the underlined word in each sentence.

22. The refugees were received at the transit camp
- A. Passage
 - B. Permanent
 - C. Temporary
 - D. Nearby
23. Abubemi is apathetic about his work
- A. Bored
 - B. Half-hearted

- C. Indifferent D. Enthusiastic
24. The man found his wife physically repellant
A. Flamboyant B. Unpleasant
C. Pleasurable D. Attractive
25. The lecturer's obscure publications counted against him during the promotion exercise.
A. Restricted B. Closed
C. Circulated D. Widespread

APTITUDE TEST

1. IT stands for _____. A. Information Technology
B. Integrated Technology
C. Intelligent Technology
D. Interesting Technology
2. A collection of related information sorted and dealt with as a unit is a _____.
A. Disk B. Data
C. File D. Floppy
3. _____ is the process of carrying out commands.
A. Fetching B. Storing
C. Executing D. Decoding
4. Promotion in marketing means _____.
A. Passing an examination
B. Elevation from one grade to another
C. Selling the products through various means
D. Selling the product in specific areas
5. Which part of the computer displays the work done?
A. RAM B. Printer
C. Monitor D. ROM
6. If the digits in the number 86435192 are arranged in ascending order, what will be the difference between the digits which are second from the right and fourth from the left in the new arrangement?
A. One B. Two C. Three D. Four
7. Each vowel of the word ADJECTIVE is substituted with the next letter of the English alphabetical series, and each consonant is substituted with the letter preceding it. How many vowel are present in the new arrangement?
A. None B. One
C. Two D. Three

8. How many such pairs of letters are there in word ENGLISH, each of which has as many letters between its two letters as there are between them in the English alphabets?
A. None B. One
C. Two D. More than three
9. In a certain code 'na pa ka so' means 'birds fly very high', 'ri so la pa' means 'birds are very beautiful' and 'ti me ka bo' means 'the parrots could fly'. Which of the following is the code for 'high' in that language?
A. Na B. Ka C. Bo D. So
10. If five is subtracted from each of the numbers, which of the following numbers will be the difference between the second digit of second highest number and the second digit of the highest number?
A. Zero B. 3 C. 1 D. 4

In each of these question a group of letters is given followed by four combinations of number/symbol **lettered** (A), (B), (C) and (D). Letters are to be coded as per the scheme and conditions given below. You have to find out the serial letter of the combination, which represents the letter group. Serial letter of that combination is your answer **letters # Q M S I N G D K A L P R B J E Number /symbol #7 @ 4#% \$ 612E5* 983 Conditions:**

- (i) if the first letter is a consonant and the last a vowel, both are to be coded as the code of the vowel.
- (ii) if the first letter is a vowel and the last a consonant, the codes for the first and the last are to be interchanged
- (iii) if no vowel is present in the group of letters, the second and the fifth letters are to be coded as ©.
11. **BKGQJN**
A. 9© \$7©% B. ©9\$7%©
C. 91\$78% D. %1\$789
12. **IJBRLG**
A. #89*£\$ B. #89*£#
C. \$89*£# D. \$89*£\$
13. **DMBNIA**
A. 6@9%#2 B. 2@9%
C. 2@9%@2 D. 2©9%@2
14. One monkey climbs a pole at the rate of 6mts/min and fell down 3mts in the alternate. Length of the pole is 60mts, how much time it will take to reach the top?

A. 31 B. 33 C. 37 D. 40

15. X men work for X days to produce X products, then Y men can produce Y products in ____ days.

A. y^3/x^2 B. X^y/y^3
C. y^3/x^2 D. y^2/x^3

16. $\sqrt{12} - \sqrt{12} + (\sqrt{12}) + \dots \text{Infinity} = ?$
A. 5 B. 4 C. 7 D. 3

17. An athlete runs 200 metres seconds. His speed (in km/hr) is:

A. 20 B. 24
C. 28.5 D. 30

18. A train passes two bridges of lengths 800m and 400 m in 100 seconds and 60 seconds respectively. The length of the train is

A. 80m B. 90m
C. 200m D. 150m

19. Find the least multiple of 23, which when divided by 18, 21 and 24 leaves the remainder 7, 10 and 13 respectively

A. 3013 B. 3024
C. 3002 D. 3036

20. The size of the bucket is N kb. The bucket fill at the rate of 0.1 kb per millisecond. A programmer sends a program to receiver. There it waits for 10 milliseconds. And response will be back to programmer in 20 milliseconds. How much time the program takes to get a response back to the programmer, after it is sent?

A. 20ms B. 40ms
C. 18ms D. 30ms

21. A fisherman's day is rated as good if he catches 9 fishes, fair if 7 fishes and bad if 5 fishes. He catches 53 fishes in a week and had all good, fair and bad days in the week. How many good, fair and bad days did the fisherman had in the week?

A. 4 good, 1 fair and 2 bad days
B. 3 good, 2 fair and 1 bad days
C. 2 good, 3 fair and 3 bad days
D. 4 good, 3 fair and 2 bad days

22. If in each number the first and the second digits are interchanged, which will be the third highest number?

A. 489 B. 654 C. 953 D. 783

23. **Antonyms:** equanimity **Analogies:** celebrate: marriage:

A. Window: bedroom
B. Lument: barevement
C. Pot: pan
D. Face: penalty

24. **Antonyms:** apathetic **Analogies:** Germ: disease

A. Man: woman B. Doctor: medicine
C. War: destrusion D. Shopkeeper: goods

25. Three of the following four are alike in a certain way and so form a group. Which is the one that does not belong to that group?

A. Stem B. Tree
C. Root D. Branch

GENERAL PAPER

1. Which of these will obey ohm law when connected to the terminals of a battery?

A. Tungsten wire B. Diode
C. Resistor D. Transistor

2. To produce direct current from a dynamic we use

A. Two slip rings B. A split ring
C. One slip ring D. Two split rings

3. An electric heat rated 15v, 40watts supplies heat to a body of mass 1.5kg and specific heat capacity of 460 J/kgK for 5 minutes. Calculate the temperature rise of the body if no heat is lost.

A. 12°C B. 1.7°C
C. 1.3°C D. 17.4°C

4. The velocity of a body of mass 250kg is steadily 50km/h. Its kinetic energy is

A. 12.5KJ B. 5.0kJ
C. 25KJ D. 625KJ

5. Which of these is odd?

A. Force B. Speed
C. Acceleration D. Displacement

6. A truck of mass 259kg moves with a uniform velocity of 36km/h. Its energy is

A. 9,000J B. 162KJ
C. 324KJ D. 648KJ

7. Mirage is as a result of
A. Total reflection
B. Total internal refraction
C. Total internal
D. Mono reflection
8. Impedance is the resistance offered by
A. Resistor
B. Indicator
C. Capacitor
D. Inductor and capacitor combined
9. Calculate the resonance frequency of a circuit with a 4H inductance coil and 2uF capacitor in series
A. 56.0Hz B. 0.018HZ
C. 0.02HED D. 100HE
10. When two plan mirrors are inclined at angle of 60° the number of images formed is
A.3 B.5 C.6 D.11
11. Which of these can convert sound energy to electrical energy
A. Loudspeaker B. Telephone car piece
C. AIC generator D. Microphone
12. A 50kg mass is lifted up 5 metres in 5 seconds. If acceleration due to gravity is 10m/s², the power utilized is
A. 200w B.500w
C.250w D. 100w
13. A stone at the bed of a stream which is 7.5m deep is viewed by a boy. If the refractive index of the stream is 1.33, how close to the surface of the stream will the stone apparently be.
A. 6.17m B. 8.83m
C. 2.82m D.5.64m
14. If the electronic configuration of an element is 1S² 2S² 2P⁵, how many unpaired electrons are there?
A.1 B.2 C.3 D.5
15. Which of the following alloys contain iron?
A. Steel and permalloy
B. Duralumin and steel
C. Soft solder and duralumin
D. Brass and bronze
16. The density of a certain gas is 1.98gdm³ at s.t.p. What is the molecular mass of the gas?
A. 26.og B. 31.0g
C. 54.0g D. 44.0g
(Molar volume of a gas at s.t.p. 22.4dm³)
17. The oxidation state of chlorine in HClO₄, is
A. -1 B. -5 C. +7 D. +1
18. A hydrocarbon x with a molecular mass of 26 consist of 92.3% carbon. What is its molecular formula?
A. C₂H₂ B. C₃H₃
C. C₃H₃ D.C₅H₅
19. What volume of 11.0m hydrochloric acid must be diluted to obtain 1dm³ of 0.05m acid?
A. 0.05dm³ B. 0.10dm³
C. 0.55dm³ D. 11.0dm³
20. Which of the following physical properties decreases down a group in the periodic table?
A. Atomic radius B. Ionic radius
C. Electropositivity D. Electronegativity
21. Aromatic and aliphatic hydrocarbons can be distinguished from each other by the
A. Action of bromine
B. Use of polymerization reaction
C. Action of heat
D. Use of oxidation reaction
22. If 30cm³ of oxygen diffuses through a porous plug in 7s, how long will it take 60cm³ of chlorine to diffuse
A. 12s B. 14s C.21s D. 30s
(O = 16, Cl=35.3)
23. Which of the following compounds is a soapless detergent?
A. C₁₂H₂₅OSO₃Na B. C₁₇H₃₅COONa
C. C₆H₅OH D. C₄H₉COOC₂
24. In order to remove one electron from a orbital of gaseous sodium atom, about 495kj mol⁻¹ of energy is required. This energy is referred to as
A. Electron affinity B. Ionization energy
C. Activation energy D. Electronegativity
25. The entropy and enthalpy of a system arc a measure of

- A. Degree of disorderliness and heat content
→ respectively
B. Heat content and degree of disorderliness
→ respectively
C. Heat content of a system only
D. Degree of disorderliness only
26. 0.46g of ethanol when burned raised the temperature of 50g of water by 14.3k. Calculate the heat of combustion of ethanol
A. +3000kJmol⁻¹ B. +3000kJmol⁻¹
C. -300kJmol⁻¹ D. -300kJmol⁻¹
27. Given $11001_{\text{two}} = 2x + 1$, find the value of x where two stands for binary
A.12 B.5 C. 16 D.8
28. The quadratic equation whose roots are -2 and -3 is
A. $2x^2+x-6=0$ B. $x^2+5x+6=0$
C. $x^2+x-6 = 0$ D. $x^2-2x-3 = 0$
29. Simplify $\log_2 6 - \log_2 4 + 2\log_2 4$
A.-1 B.6 C.4 D.5
30. If $y = \cos x$, then $\frac{dy}{dx}$ is
A. Sin x B. Tan x
C. -Cos x D. -Sin x
31. A polygon which has at least one angle greater than 180° is called
A. Convex B. Regular
C. Reflex D. Re-entrant
32. One interior angle of a convex polygon that is regular is 160°. How many sides has the polygon?
A.12 B.8 C.15 D.18
33. If 15, x, y, 27 are terms in an AP then the value of 2(x+y) is
A. 84 B. 36 C. 50 D. 48
34. The total surface area of a spherical axis 7cm where $\pi = \frac{22}{7}$ is
A. 616cm² B. 418cm²
C. 364cm² D. 512cm²
35. Find the remainder when $x^3 - 6x + 1$ is divided by $x-3$
A. 12 B. 16 C. 10 D. 14
36. Simplify $\frac{\sqrt[3]{2-1}}{\sqrt{2-1}}$
A. $\frac{\sqrt{2-1}}{3}$ B. $\frac{-}{\sqrt{2-1}}$
C. $2\sqrt{2} + 3$ D. $3\sqrt{2} + 1$
37. At what rate can N6560 yields a simple interest of N240 in 4 years
A. 9.2% B. 13.3%
C. 12.4% D. 16.6%
38. The sum of the interior angles of a polygon is 1260°. How many sides has the polygon?
A. 16 B. 22 C. 9 D.30
39. In unicellular organisms, essential nutrients can be transported directly to all part of their body by the process of diffusion only because unicellular organisms have
A. Outer membrane made of cellulose
B. A large surface area to volume ratio
C. A large volume to surface area ration
D. Their body always immersed in the nutrient material
40. Which of the following is the function of the hormone in the is let of langerhans?
A. It is used in protein metabolism
B. It is involved in sugar synthesis
C. It aids the rate at which glucose is converted to glycogen
D. controls the fluid content of the body
41. Which of the following statement is not correct about the ecological effect of agricultural activities?
A. The use of herbicides and pesticides in farms can cause pollution.
B. Bush-burning encourages decertification
C. Excessive use of fertilizers in farm may have harmful effect.
D. Monocropping is one of the cultural method of controlling pest.
42. Which of the following statement about the transfer of energy in an ecosystem is correct?
A. Energy increases at higher tropic level
B. More organisms can be supported at higher levels.

- C. There are fewer organisms at lower trophic levels.
D. Energy is lost at each trophic level.
43. A vaccine is introduced into a person's body to
A. Destroy all disease-causing organism in the body
B. Stimulate the body to produce antibodies
C. Enhance the production of red blood cells
D. Isolate disease causing pathogen from the blood
44. Which. of the following statement about phenylthiocarbamide (PTC) is not correct? A. Some people find the taste of PTC bitter 1. Others find the substance to be tasteless
B. Others find the substance to be tasteless
C. Ability to taste this chemical is inheritable
D. Ability to taste this chemical is not heritable
45. The order of evolutionary trend in plants is A. Bryophyta Thallopiphyta, Pteridophyta and Spetmatophyta
B. Thallopiphyta, Bryophytta, Pterydophyta and Spermatophyta
C. Spermatophyta, Pterydophyta, Bryophyta and Thalophyta
D. Bryophyta, Pteridophyta, Spermatophyte and Thallphyta
46. Which of the following statements about the circulation of blood is not correct?
A. Deoxygenated blood flows in the heart through the VNA cavae
B. Blood is pumped out of the heart through the aorta
C. Oxygenated blood from the lungs is carried to the left auricle
D. The oxygenated blood enters the lungs through the pulmonary vein

With the information provided below on photosynthesis answer questions 47 and 48 (1) CO₂ (ii) Chlorophyll (iii) water (v) AT? (v) NADPH

47. Which of the above factors can be a limiting factor for the process to proceed? A. i and ii only
B. i and iii only
C. i and iv only D. i, iii and iv only
48. The factors required for the dark phase of photosynthesis are

- A. iii and iv only B. iii and v only
C. iii, iv and v only D. iv and v only

49. A potometer was used to compare the rate of loss of the same volume of water from a plant under different environmental conditions namely: in the airy laboratory, in the cupboard, under fan, under sun and at the seashore. Which of the readings below was likely obtained in a cupboard?
A. 15 sees B. 30 sees
C. 25 sees D. 20sccs
50. Atmospheric oxygen get to the mammalian blood stream through the following paths indicated below (i) mouth (ii) trachea (iii) bronchile (iv) nostril (v) bronchus (vi) alveoli. Indicate the correct pathway
A. → → → → iii v vi iv B. ii
→ → → → iv v vi iii
C. iv → → → → ii viii vi D. ii
→ → → → i vi v iii

SOLUTION

Eng.

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. C | 2. A | 3. A | 4. A | 5. C | 6. C | 7. B |
| 8. C | 9. C | 10. D | 11. A | 12. C | 13. A | 14. A |
| 15. A | 16. D | 17. A | 18. A | 19. D | 20. C | 21. A |
| 22. B | 23. D | 24. D | 25. D | | | |

Apt. Test

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. B | 3. C | 4. D | 5. C | 6. D | 7. A |
| 8. A | 9. D | 10. A | 11. B | 12. C | 13. B | 14. D |
| 15. B | 16. B | 17. D | 18. A | 19. A | 20. A | 21. D |
| 22. C | 23. D | 24. B | 25. B | | | |

Gen. Paper

- | | | | | | | |
|-------|-------|-------|-------|-------|-------|-------|
| 1. A | 2. D | 3. D | 4. C | 5. B | 6. | 7. B |
| 8. D | 9. A | 10. B | 11. A | 12. | 13. D | 14. A |
| 15. A | 16. D | 17. C | 18. A | 19. A | 20. D | 21. A |
| 22. C | 23. A | 24. B | 25. A | 25. C | 27. A | 28. B |
| 29. B | 30. D | 31. D | 32. D | 33. A | 34. A | 35. C |
| 36. D | 37. A | 38. C | 39. B | 40. C | 41. D | 42. D |
| 43. B | 44. D | 45. B | 46. D | 47. B | 48. D | 49. B |
| 50. C | | | | | | |

**POST-JAMB SCREENING TEST
REVISION QUESTIONS AND ANSWERS
2010/2011 SESSION DELSU 2019
AFTERNOON SESSION**

ENGLISH

Choose the Word that is opposite in meaning to the underlined word in each sentence

1. It is generally believed that misers are not loved by many
A. Spender B Hoarders
C. Savers D. Spendthrifts
2. Ese and her brother are simply munificent
A. Angry B. Niggardly
C. Wealthy D. Happy
3. He is known for making threadbare arguments as a scholar
A. Justifiable B, Baseless
C. Empty D. Militant

Read each passage and answer the questions 4 -8 choosing

A British electronic manufacturer has built a factory in which quartz crystals can be grown in 3 weeks. The natural growth of quartz takes 3 million years. The manufactured crystals are exceptionally pure and this is important because quartz, in transistors is used in an astonishing number of devices where constant television transmitters and receivers, space satellites and computers are familiar examples. In order to make artificial quartz, very small pieces of the natural crystal imported from Brazil are placed in long, narrow, steel cylinders. A high pressure and a high temperature are maintained within the cylinders for 3 weeks and at the end of this time, the small fragments have grown to the required weight of one pound.

4. Artificial quartz is produced
A. In Brazil
B. From tiny transistors
C. From natural crystals with steel cylinders
D. In an astonishing number of devices
5. Artificial quartz as compared with natural is
A. Superior B. Inferior
C. Varied D. Limited in use
6. According to the passage, quartz is used
A. Only in a few devices
B. In a lot of devices

- C. In wrist watches
- D. For steel cylinders

7. Transistors have
A. Helped to produce quartz
B. Retarded progress in electronics
C. Made transistors and receivers smaller and lighter
D. Made transmitters and receivers bigger and heavier
8. A suitable title for the passage is
A. Natural quartz
B. Receivers
C. Manufactured and natural quartz crystals
D. Manufacture of transistors, satellite and computers

Complete each of the following sentences by choosing the option that most suitably fills the space

9. If I went to the cinema I _____ early
A. Am returning B. Had returned
C. Would return D. Shall return
10. Last week _____ at the theatre was the largest I had ever seen
A. Audience B. Congregation
C. Assembly D. Convention
11. My father cannot bear the _____ of the concoction that emanates from my mother's soup pot
A. Savouring B. Smell
C. Aroma D. Scent
12. If this had happened to you, what _____
A. Would you do
B. Would you have done
C. Have you done
D. Would you be doing
13. The governor commended the society's _____ service to the nation
A. Humanistic B. Humanitarian
C. Humanity D. Human
14. If we went to a European country _____ Britain, we should need a substantial amount of money to pay our way
A. Let us say B. Like say
C. Like D. As an example

Choose the option that best conveys the meaning of the underlined portion of each of the following sentences

15. We all have both good and bad characteristics. Either is, however easily manifested in time of crisis
A. Demanded B. Demonstrated
C. Highlighted D. Rediculed
16. Ese takes his studies lightly because he is a discoaddict
A. Carefully B. Unseriously
C. Gloomily D. Humorously
17. Can you give a good explanation to your behaviour?
A. Clear up B. Account for
C. Give up D. Makeup
18. All the efforts made to settle the quarrel were futile
A. Fruitless B. Unreasonable
C. Successful D. Remarkable
19. My son is naturally taciturn A. Loquacious
B. Dumb
C. Lively D. Friendly
20. Accountability is certainly a desirable quality in a politician
A. Respectability B. Diligence
C. Responsibility D. Courage
21. Due to the fact that we are now insolvent, the bill has to wait
A. Bankrupt B. Unsettled
C. Overworked D. Affluent
22. My uncle thinks that her daughter could be easily taken in by her husband's pretences
A. Blackmailed B. Flattered
C. Deceived D. Enamoured
23. The man has a big heart but he is inept in following witty conversation
A. Large B. Warm
C. Huge D. Fragile
24. For a priest to be successful, I should, from time to time, review his action
A. Examine B. Inspect
C. Stare at D. Visualize

25. My sister used her savings to bring up her niece
A. Support B. Lead
C. Accompany D. Feed

APTITUDE TEST

1. The average age of a class is 15.8 years. The average age of boys in the class is 16.4 years while that of girls is 15.4 years. What is the ratio of boys to girls in the class?
A. 1:2 B. 3:4 C. 3:5 D. 2:3
2. If one-seventh of a number exceeds its eleventh part by 100 then the number is
A. 7,700 B. 1100
C. 1925 D. 1825
3. Gold is 19 times as heavy as water and copper is 9 times as heavy as water. In what ratio should these be mixed to get an alloy 15 times as heavy as water?
A. 1:1 B. 2:3 C. 1:2 D. 3:2
4. A triangle always has exactly
A. One acute angle
B. At least two acute angle
C. Exactly one right angle
D. Exactly two acute angles
5. Pipe A can fill in 20 minutes and Pipe B in 30 mins and Pipe C can empty the same in 40 mins. If all of them work together, find the time taken to fill the tank
A. 1/7mins B. 172/7mins
C. 71/9mins D. 121/5mins
6. The smallest number which when diminished by 3 is divisible by 21, 28, 36 and 45 is
A. 869 B. 1263 C. 4320 D. 427
7. The process of a computer receiving information from a server on the internet is known as ____
A. Pulling B. Pushing
C. Downloading D. Transferring
8. The role of ____ generally is to determine a buyer's needs and match it to the correct hardware and software.
A. Computer scientist
B. Computer sales representative
C. Computer consultant
D. Corporate trainer

9. A (n)_____ contains commands that can be selected
 A. Pointer B. Menu
 C. Icon D. Button
10. Sending an e-mail is similar to_____ A.
 Picturing an event B. Narrating a story
 C. Writing a letter D. Creating a drawing
11. A series of instructions that tells a computer what to do and how to do it is called a____
 A. Program B. Command
 C. User response D. Processor

Rearrange the following six sentences (1), (2), (3), (4), (5), and (6) in the proper sequence to form a meaningful paragraph; then answer the questions given below them.

- (1) She eased out something from her waist folds
 - (2) Nnena welcomed a customer early in the day.
 - (3) Once visible, Nnena realized it was a pair of gold bangles.
 - (4) The woman carefully counted the money and then left.
 - (5) He took the bangles and placed some money in the woman's palm.
 - (6) This customer was a peasant woman wearing a discoloured sari and old anklets.
12. Which of the following should be the **FIRST** sentence after rearrangement?
 A.1 B.2 C.3 D.4
13. Which of the following should be the **SECOND** sentence after rearrangement?
 A.2 B.3 C.4 D.4
14. Which of the following should be the **THIRD** sentence after rearrangement?
 A.1 B.2 C.3 D.6
15. Two numbers are respectively 25% and 20% less than a third number. What percent is the first number of the second.
 A. 5.85% B. 75.85%
 C. 80.60% D. 93.5%
16. A can do a piece of work in 4 hours. B and C can do it in 3 hours. A and C can do it in two hours. How long will B take to do it?
 A. 10 hours B. 12hours
 C. 8 hours D. 24hours

17. A face of the clock is divided into three parts. First part hours total is equal to the sum of the second and third part. What is the total of hours in the bigger part?
 A. 6 hrs B. 7 hrs
 C. 4hrs D. 5 hrs
18. 2 trees are there. One grows at $\frac{3}{5}$ of the other. in 4 years total growth of the trees is 8ft. What growth will smaller tree have in 2 years.
 A. 1.2 B. 1.4 C. 1.1 D. 1.5
19. Joseph finishes a work in '7 days, Nicholas finishes the same job in 8 days and Edwin in 6 days. They take turns to finish the work. Joseph on the first day, Nicholas on the second and Edwin on the third day and then Joseph again and so on. On which day will the work get over?
 A. 3rd B. 6th C. 9th D. 7th
20. A man, a women and a child can do a piece of work in 6 days, man can do it in 14 days, women can do it 16 days, and in how many days can a child do the same work?
 A. 24 days B. 25days
 C. 16 days D. 20days

Which of the phrases (A), (B) and (C) given below each sentence should replace the phrase printed in **bold** in the sentence to make it grammatically correct? If the sentence is correct as it is given and 'No correction is required', mark (D) as the answer.

21. I woke up early in the morning and had a **steamer** cup of coffee A. Has a steamer B. Has a steaming
 C. Had a steaming
 D. No correction required
22. Tired of **being harassed** by the goons, I finally called the police and complained_____
 A. In being harassed
 B. Of being harass
 C. In be harassed
 D. No correction required
23. The ratio of incomes of C and D is 3:4 the ratio of their expenditures is 4:5. Find the ratio of their savings if the saving of C is one Garth of his income?
 A. 11:12 B. 12:19
 C. 10:13 D. 13:15

24. If $G(0)=-1, G(1)=1$ and $G(N)=G(N-1)-G(N-2)$ then what is the value of $G(6)$?
A. 1 B. 2 C. -1 D. -2
25. A person had to multiply two numbers. Instead of multiplying by 35, he multiplied by 53 and the product was up by 540. What was the raised product?
A. 780 B. 1040
C. 1590 D. 1720

GENERAL PAPER

1. 1. 0.46g of ethanol when burned raised the temperature of 50g of water by 14.3k. Calculate the heat of combustion of ethanol
A. +3000kJmol⁻¹ B. +300kJmol⁻¹
C. -3000 kJmol⁻¹ D. -300kJmol⁻¹
2. Make u the subject of the formula $\frac{1}{v} + \frac{1}{u} - \frac{1}{f}$
A. $vf + u$ B. $\frac{1}{uvf}$
C. $\frac{uv-f}{v}$ D. $\frac{-vf}{f-v}$
3. Evaluate $\int x dx$
A. $2x + c$ B. $x + c$
C. $\frac{1x^3}{4} + c$ D. $\frac{1}{2} x^2 + c$
4. Differentiate $y = x$ with respect to x
A. $\frac{1}{6}$ B. 8 C. 4 D. $\frac{1}{4}$
5. The angle subtended by a diameter of a circle at the circumference is a/an
A. Equilateral angle B. Supplementary angle
C. Right angle D. Reflex angle
6. In two parallel sides of a trapezium are 10cm and 6cm. its height is 4cm. Find the area of the trapezium
A. 25cm² B. 32cm²
C. 40cm² D. 56cm²
7. The exterior angle of a triangle is 76°. If one of the opposite interior angles is 42° find the other opposite interior angle
A. 34° B. 52° C. 48° D. 39°
8. A trader bought a bicycle for ₦645 and sold for ₦862, Find the percentage gain
A. 41% B. 33.6%
C. 27.2% D. 51.2%

9. The speed of the flow water in a river is faster in the middle than along the bank due to
A. Differences in turbidity at different parts of the river
B. The abundance of fishes and other organisms along the bank
C. Reduced force of gravity in the middle portion of the river
D. Resistance offered by the wall of the bank.
10. If red is dominant to white and homozygous red pea-plants are crossed with homozygous white pea plants, the first filial generation will have
A. Two red flower, two white flowers
B. Four red flowers
C. Three red, one white
D. Four pink flowers
11. A population is defined as the collection of
A. Similar organism that are found in the same habitat
B. Similar organism that breed in the same habitat
C. Similar organisms that eat the same food
D. Similar organism, that interbred freely in the same habitat
12. The equation that can be used to summarize the process of anaerobic breakdown of sugar is:
A. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2$
B. $6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$ C. $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{Energy}$
D. $C_6H_{12}O_6 \rightarrow 2C_2H_5OH + 2CO_2 + \text{Energy}$
13. In animal evolution which of the following set of organisms represents the correct trend in terms of structural organization 1. Mollusca 2. Platyhelminthes
3. Nematoda 4. Protozoa
A. 4 → 1 → 2 → 3 B. 4 → 3 → 2 → 1
C. 4 → 2 → 1 → 3 D. 4 → 2 → 3 → 1
14. Which of the following is not a feature of Meiosis
A. Formation of four haploid cells
B. Two successive nuclear cell division
C. Pairing of homologous chromosomes at prophase
D. Formation of two diploid cells

15. Which of the following is a means measuring primary production? A. Rate of reproduction of animals
B. Life span of an organism
C. Number of predators in an environment
D. Harvested plants in a unit area
16. What is the percentage by mass of oxygen in $\text{Al}_2(\text{SO}_4)_3 \cdot 2\text{H}_2\text{O}$?
A. 59.25% B. 50.79%
C. 25.39% D. 14.29A%
(Al=27, S=32, H=1, O=16)
17. A balanced chemical equation obeys the law of
A. Conversation of mass
B. Multiple proportion
C. Definite proportions
D. Conservation of mass
18. To what temperature must a gas at 273k be heated in order to double both its volume and pressure?
A. 298k B. 546k
C. 819k D. 1092k
19. Neutralization involves a reaction between H_2O and
A. Cl^- B. OH^- C. NO_3^- D. CO_3^{2-}
20. In the extraction of iron in the blast furnace, limestone is used to
A. Release CO_2 for the reaction
B. Reduce the iron ore
C. Increase the strength of the iron
D. Remove impurities
21. A pin is placed under a transparent glass block of height 12cm. If the refractive index of the glass is 1.56, the image will appear to be at
22. Hypermetropia (long sightedness) is corrected by
A. Concave mirror B. Plane mirror
C. Plans concave mirror D. Convex mirror
23. Resonance in circuits is a conduction in which the circuit.
A. Produces highest resistance or impedance
B. Produces no current
C. Produces maximum current
- D. Refuses to function
24. A pinhole camera of length 12cm has the image of an object 15m away on its screen at the 12cm mark. If the height of the object is 2 metres, calculate the height of the image.
A. 2.5cm B. 3.6cm
C. 1.6cm D. 0.63cm
25. According to ____law, the induced EMF is directly proportioned to the rate of change of magnetic flux
A. Lenz's B. Faraday's
C. Ampere's D. Ohm's
26. Which of the following does not happen during inspiration?
A. Intercostals muscles contract
B. Ribs move forward
C. Thoracic cavity increase
D. Diaphragm relaxes
27. A potometer was used to compare the rate of loss of the same volume of water from a plant under different environmental conditions namely; in the airy laboratory, in the cupboard. under fan, under sun and at the seashore which of the reading below was likely obtained in a cupboard?
A. 15secs B. 30secs C. 25secs D. 20secs



Figure 1
Use figure 1 to answer questions 28 and 29

28. The testa
A. Is normally absent
B. Is included in the part numbered 1
C. Is fused with the part number 3
D. Is the part number 1
29. Reserved food is stored in the part numbered
A. 2 and 3 B. 3 only
C. 4 and 5 D. 5only
30. When an operculate fish breathes, it takes in water through the
A. Mouth and passes it out through the same way
B. Mouth and passes it out through this operculum

- C. Operculum and passes it out through the mouth
D. Nostril and passes it out through the operculum
31. A _____ can store charges until when required for use
A. Galvanometer B. Comparator
C. Diode D. Capacitor
32. The wheat stone bridge circuit usually has sets of resistors
A Five B. Two
C. Three D. Four
33. The temperature of water in a flask is 50°C . In the fahrenheit scale, this will read
A. 90°F B. 122°F
C. 27.8°F D. 60°F
34. 20cm of gas was maintained at 5°C and 760mm Hg pressure what volume of this gas will be maintained at a temperature of 35° and 800mmHg pressure
A. 20.7cm^3 B. 39.5cm^3
C. 21.05cm^3 D. 67.8cm^3
35. _____ means bending of light rays from their path
A. Reflection B. Conduction
C. Refraction D. Transmission
36. Find the 40th term of the A.P. 8,5,2
A. -407 B. -316
C. -109 D. -241
37. Find the area of a triangle of sides 3cm, 4cm and 5cm
A. 12cm^2 B. 6cm^2
C. 10cm^2 D. 8cm^2
38. Find the value of k if $(x+1)$ is a factor of x^3+kx^2+3x-2
A.5 B.12 C.-10 D.6
39. Simplify $\sqrt{405}$
A. $5\sqrt{3}$ B. $6\sqrt{2}$ C. $9\sqrt{5}$ D. $3\sqrt{6}$
40. If 3 is a root of the equation $x^2-kx+42=0$. The value of k is
A. 18 B. 19 C. 22 D. 17
41. The period of an oscillating body is 0.08sec. Its frequency is
A. 50HZ B. 78.6HZ
C. 12.5HZ D. 70HZ
42. If the force on a charge of 0.3 coulomb in an electric field is 4N, then the electric field intensity of the field is
A. 13.3N B. 0.75N
C. 3.7N D. 1.2N
43. A radio station transmits with a frequency of 2500kHz, the wave length of its waves is (take $c = 3 \times 10^8 \text{m/s}$)
A. 120m B. 83.3m
C. 7500m D. 2000m
44. Which of the following are structural isomers?
A. Propanoic acid and propan-2, 2-diol
B. Ethanoic acid
C. Ethane-1, 2-diol and ethanoic acid
D. Propanal and propanone
45. When a solid substance disappears completely as a gas on heating, the substance is said to have undergone
A. Crystallization B. Evaporation
C. Evaporation D. Distillation
46. Which of the aqueous solutions with the pH values below will liberate hydrogen when it reacts with magnesium metal? A. 12.0
B.7.0 C. 6.5 D. 3.0
47. Which of the following electron transitions results in the emission of energy
A. 3p to 3s B. 3p to 4p
C. 2s to 2p D. 1s to 2s
48. The fundamental difference between the three states of matters is the
A. Shape of their particles
B. Number of particles in each state
C. Shape of the container they occupy
D. Degree of movement of their particles
49. A particle that contains 9 protons, 10 neutrons and 10 electrons is a
A. Positive ion
B. Neutral atom of metal
C. Neutral atom of a non-metal
D. Negative ion

50. When sodium reacts with water, the resulting solution is
- A. Alkaline B. Acidic
C. Neutral D. Weakly acidic

SOLUTION Eng.

1. A 2. B 3. A 4. C 5. A 6. B 7. C
8. C 9. C 10. A 11. C 12. B 13. B 14. C
15. B 16. B 17. B 18. A 19. B 20. C 21. A
22. C 23. A 24. A 25. A

Apt. Test

1. A 2. A 3. D 4. B 5. C 6. B 7. C
8. C 9. C 10. C 11. B 12. B 13. D 14. A
15. A 16. C 17. A 18. A 19. C 20. D 21. C
22. D 23. B 24. A 25. C

Gen. Paper

1. C 2. D 3. D 4. 5. C 6. B 7. A
8. B 9. A 10. B 11. D 12. D 13. B 14. D
15. D 16. A 17. D 18. D 19. B 20. D 21. C
22. D 23. C 24. C 25. A 26. A 27. B 28. D
29. A 30. B 31. D 32. D 33. B 34. B 35. C
36. C 37. B 38. D 39. C 40. D 41. C 42. A
43. A 44. A 45. C 46. D 47. D 48. D 49. D
50. A